

Sometimes-Competing Retrieval (SOCR): A Formalization of the Comparator Hypothesis

Steven C. Stout
Jacksonville State University

Ralph R. Miller
Binghamton University, State University of New York

Cue competition is one of the most studied phenomena in associative learning. However, a theoretical disagreement has long stood over whether it reflects a learning or performance deficit. The comparator hypothesis, a model of expression of Pavlovian associations, posits that *learning* is not subject to competition but that *performance* reflects a complex interaction of encoded associative strengths. That is, subjects respond to a cue to the degree that it signals a *change* in the likelihood or magnitude of reinforcement relative to that in the cue's absence. Initially, this performance-focused view was supported by studies showing that posttraining revaluation of a competing cue often influences responding to the target cue. However, recently developed learning-focused accounts of retrospective revaluation have revitalized the debate concerning cue competition. Further complicating the picture are phenomena of cue facilitation, which have been addressed less frequently than cue competition by formal models of conditioning of either class. The authors present a formalization and extension of the comparator hypothesis, which results in sharpened differentiation between it and the new learning-focused models.

Keywords: Pavlovian conditioning models, compound conditioning, retrospective revaluation, cue competition, second-order conditioning

At its inception, the field of associative learning was concerned with how responding to a single target predictor of a biologically important outcome is affected by numerous controlling variables. In the late 1960s and early 1970s, the focus of the field shifted to the study of how responding is affected by the interaction between multiple predictor cues. Earlier, Pavlov (1927) had demonstrated that multiple cues compete in their control over behavior. Specifically, he observed stimulus control by a target cue, or conditioned stimulus (CS), to be attenuated if it was paired with the outcome, or unconditioned stimulus (US), in the presence of another cue, a phenomenon he called *overshadowing*. However, it was Kamin's (1968, 1969) discovery of a similar phenomenon, *blocking*, that renewed interest in cue competition and led to much research and theorizing. The blocking effect refers to decreased behavioral control by a target cue if it is paired with the outcome in the presence of a previously established predictor of the outcome. Similarly, Rescorla (1968) reported that conditioned responding is impaired when unsignaled outcomes are added to the intervals between the cue–outcome pairings (i.e., the *degraded contingency*

effect), leading to competition between the context and the target CS. At about the same time, the phenomenon of Pavlovian *conditioned inhibition* received renewed empirical interest (Rescorla, 1969). Conditioned inhibition occurs when reinforcement of one cue (an *excitor*) is juxtaposed with nonreinforcement of that cue in compound with the target (inhibitory) cue. Inhibitory cues control response tendencies opposite to those of excitors, they are more difficult to condition as excitors, and they reduce conditioned responding to other excitatory cues. Inhibition can also be viewed as a form of cue competition (perhaps better called *cue interaction* in this case, Chapman, 1991), if the dimension of behavioral control is envisioned to range from inhibition to excitation. Inhibitory and excitatory cues compete in the sense that their response potentials are inversely correlated.

The seminal interpretation of cue competition, including conditioned inhibition, was offered by Kamin (1968, 1969) in his explanation of the blocking effect. It was later formalized and extended by Rescorla and Wagner (1972) in their highly influential model (R-W; for reviews, see Miller, Barnet, & Grahame, 1995; Siegel & Allan, 1996), which is to date one of the most referenced papers in psychology and the most commonly explicated formal model described in learning textbooks. According to this view, animals form an association between a paired cue and outcome to the extent that all currently present cues fail to predict the outcome. The model can be formalized with the following equation for the acquisition of associative strength:

$$\Delta V_{X,O} = \alpha_X * \beta_O (\lambda_O - \Sigma V_i). \quad (1)$$

$\Delta V_{X,O}$ stands for the change in the associative strength (V) of cue X as a result of its being paired with the outcome on trial n (i.e., $\Delta V_{X,O} = V_{X,O}^n - V_{X,O}^{n-1}$). The learning-rate parameters, $\alpha_{X,O}$ and β_O , represent the associability of cue X and the outcome, respectively. λ_O represents the maximum associative strength support-

Steven C. Stout, Department of Psychology, Jacksonville State University; Ralph R. Miller, Department of Psychology, Binghamton University, State University of New York.

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Correspondence concerning this article should be addressed to Ralph R. Miller, Department of Psychology, Binghamton University, State University of New York, Binghamton, NY 13902-6000. E-mail: rmiller@binghamton.edu

able by a given outcome on a single trial ($\lambda > 0$ for a presented outcome, and $\lambda = 0$ for an omitted outcome), and $\Sigma V_{i,O}$ is the sum of the associative strengths with respect to the outcome of all stimuli (indexed i), including cue X, present on that trial. Thus, changes in the associative strength of cue X ($\Delta V_{X,O}$) are proportional to the discrepancy between the outcome actually presented on a trial and the expectation of the outcome on the basis of all the stimuli present on that trial ($\lambda_O - \Sigma V_{i,O}$). Other cues present in the situation change associative strength according to an equation of the same form.

The R-W model is conceptually elegant and effective in accounting for the aforementioned cue-competition effects. It views blocking as the result of the competing cue (the blocking stimulus, A) having already been established as a predictor of the outcome prior to the compound pairings of the blocking and blocked cues with the outcome (i.e., AX–outcome). This leaves a small discrepancy between the outcome expected and the outcome presented, supporting further acquisition of an association to the blocked target cue. The adverse effect of degraded contingency treatment on conditioned responding (Rescorla, 1968) is explained in a manner analogous to blocking, except that the context of conditioning, rather than a nontarget punctate cue, serves as the blocking stimulus. Similarly, conditioned inhibitory cues develop negative associative strength through the discrepancy between the expected outcome (expected because the associated cue has been made excitatory through previous pairings with the outcome) and the absence of a delivered outcome. Thus, rather than predicting the presence of the outcome, conditioned inhibitors predict the absence of the outcome. Conditioned inhibition is a form of cue competition in the sense that the associative value of the target cue is driven negative by the positively valued excitatory companion cue; that is, the two cues compete to predict the surprising non-occurrence of the outcome.

After the appearance of the R-W model, rivals arose that relied on mechanisms other than competition for associative strength to account for cue competition. For example, Mackintosh (1975) proposed that cue competition results when different predictors are compared in their effectiveness in predicting the outcome. Attention to less predictive cues decreases, whereas attention to more predictive cues increases; that is, cue competition arises from competition for attention. Conversely, Pearce and Hall (1980) formulated an account of cue competition in which a predictor loses associability when its consequences are effectively predicted by all cues present on that trial. Wagner (1981) viewed it as competition between dynamic excitatory and inhibitory tendencies. Despite these differences, these models, and all other models that focus on acquisition processes, share a fundamental assumption—that conditioned responding to a target cue is a monotonic transformation of that cue's, and only that cue's, associative strength. In this view, any evidence of excitatory response deficits or inhibition must directly reflect target cue–outcome associations formed during acquisition. When a target cue is presented alone during a test trial, the subject is assumed to disregard all other associations formed during preceding learning experiences. This hypothesis is also used by more recent theories of associative learning that emphasize postacquisition modification of the cue–outcome association resulting from treatments of its associate without presentation of the training CS itself (e.g., Dickinson & Burke, 1996; Van Hamme & Wasserman, 1994). It might be called

the *what-you-see-is-what-you-have* (WYSIWYH) hypothesis, after its contention that during testing of a single target cue, conditioned responses directly reflect the target cue–outcome association. In some WYSIWYH models (e.g., Daly & Daly, 1982; Pearce, 1987; Pearce & Hall, 1980; Wagner, 1981), conditioned responding may result from a combination of separately conditioned excitatory and inhibitory associations between the target cue and the outcome, but the basic principle is the same: Only direct associations between the target cue presented at test and the outcome control conditioned responding.

Expressed as a response rule, the simplest WYSIWYH models (that is, those without separate inhibitory tendencies) state that responding to a single cue takes the following form:

$$R_X \approx V_{X,O}. \quad (2)$$

This equation states that the response (R) to cue X that occurs when X is presented centrally reflects the strength of the X–O association ($V_{X,O}$). Models of this nature view cue interaction treatments as modifying the strength of $V_{X,O}$.

The Comparator Hypothesis

Several years after the publication of the R-W model, the WYSIWYH hypothesis seemed to be challenged by studies demonstrating that cue-competition deficits can be alleviated through procedures that do not involve further experience with the target cue. For example, Kaufman and Bolles (1981; see also Matzel, Schachtman, & Miller, 1985) showed that posttraining extinction (so-called *associative deflation*) of an overshadowing stimulus (A) following overshadowing treatment (AX $\rightarrow$ outcome) increases (i.e., recovers) responding to the target cue (X). Additional studies obtained recovery from blocking (Arcediano, Escobar, & Matute, 2001; Blaisdell, Gunther, & Miller, 1999) and the relative stimulus validity effect (Cole, Barnet, & Miller, 1995). Other studies found a reduction in conditioned inhibition as a consequence of post-training extinction of the target cue's companion stimulus (e.g., Hallam, Grahame, Harris, & Miller, 1992; Kasprow, Schachtman, & Miller, 1987; Lysle & Fowler, 1985) and an increase in conditioned inhibition as a consequence of posttraining pairings of the target cue's companion stimulus with the outcome (so-called *associative inflation*; i.e., A–outcome pairings; Amundson, Wheeler, & Miller, 2005). Such effects have been called *empirical retrospective revaluation*. These and many other successful recovery-from-cue-interaction manipulations (reviewed by Miller, Kasprow, & Schachtman, 1986) seemed to challenge the WYSIWYH assumption that cue-interaction treatments result uniquely from direct changes in the associations between target cues and outcomes (i.e., learning). They invited interpretations in which cue-competition phenomena, including conditioned inhibition, were based on comparisons between competing associations that occurred at the time the cue was tested (i.e., they reflected performance or response rules; e.g., Denniston, Savastano, & Miller, 2001; Gallistel & Gibbon, 2000; Gibbon & Balsam, 1981 [in some formulations]; Miller & Schachtman, 1985). Models of this class have been termed *comparator models* (e.g., Mazur, 2002, p. 98).

Miller and Matzel's (1988) comparator hypothesis, the formalization and extension of which is the central focus of this paper, assumes that associations are formed through contiguity alone. These associations are formed independently of each other; that is,

there is no competition for a limited resource, such as associative strength or attention, during the learning phase. Further, the comparator hypothesis assumes that there can be no modification of the association between two cues unless at least one of those cues is physically presented. The learning component of the model is therefore essentially the same as that proposed by Bush and Mosteller (1955). However, its response rule rejects the WYSIWYH assumption and explores the extent to which it is possible to account for numerous phenomena of Pavlovian conditioning with a response rule that combines the many simple associations that a subject has learned through contiguity alone. Its rejection of the WYSIWYH hypothesis of other models from the animal learning tradition (e.g., Rescorla & Wagner, 1972) makes the comparator hypothesis more similar to interference models of human memory retrieval (particularly models of recall, recognition, and paired associate learning) in which response competition between associations for expression on test trials is a time-honored notion (e.g., Underwood, 1957).

A mathematical learning rule that captures trialwise noninteractive increases in associative strength, as proposed by Miller and Matzel (1988), is presented in Equations 3A and 3B:

$$\Delta V_{S1,S2} = s1 * s2 * (1.0 - V_{S1,S2}). \quad (3A)$$

$\Delta V_{S1,S2}$ is the trialwise increase of the association $V_{S1,S2}$ between any two arbitrary stimuli, S1 and S2, when they are conjointly presented. By association, we mean the potential of stimulus S1 to retrieve a representation of stimulus S2. The salencies of S1 and S2 are $s1$ and $s2$, respectively. As is conventional, a salience of 0 represents cue absence, and a salience value between 0 and 1 represents cue presence. The salience of a cue is assumed to be a joint function of the stimulus's physical intensity and biological significance. Equation 3B represents the trialwise decline in associative strength when S1 is presented without S2:

$$\Delta V_{S1,S2} = -k1 * s1 * V_{S1,S2}, \quad (3B)$$

where $k1$ is a decremental learning-rate parameter ($0 \leq k1 \leq 1.0$) that applies on trials on which the outcome S2 is absent, and $s1$ is the salience of stimulus S1 that is presented on the extinction trial. Notice that without further qualification, Equation 3A assumes that two stimuli presented conjointly will develop two equal unidirectional associations between them. However, in the simulations of the model, we assume only that these two associations develop when S1 and S2 are simultaneously presented. If S1 precedes stimulus S2, only $V_{S1,S2}$, and not $V_{S2,S1}$, will be formed. Equation 3B assumes that only $V_{S1,S2}$ (not $V_{S2,S1}$) will be decremented (an issue concerning which there are mixed data; e.g., Arcediano, Escobar, & Miller, 2005). Associations are assumed to be unidirectional in both acquisition and extinction. An additional consequence of these simple learning rules is that all associations are positive in value, that is, excitatory. Inhibitory behavior, which in some models reflects negatively valued associations (e.g., Rescorla & Wagner, 1972) and in other models reflects a second learning variable that is independent of the excitatory variable (e.g., Wagner, 1981), arises from the interaction of excitatory associations. Finally, our implementation of this model has no role for stimulus salience, especially US magnitude, on asymptotic responding. We left this standard component out of the model, because these learning rules are meant to apply to biologically insignificant cues

as well as biologically significant cues, and there is little evidence that stimulus salience affects asymptotic associative strength with biologically insignificant cues. Moreover, this is not a major drawback with biologically significant stimuli, as many effects of US magnitude can be attributed to its effect on acquisition rate, because without many CS-US pairings, acquisition rate and asymptote are confounded.

Consider a system that combines the noncompetitive-learning-rule Equation 3 with WYSIWYH response-rule Equation 2 (in which cue $X = S1$ and outcome $O = S2$). This system would describe an animal that learns about a cue-outcome association independently of other stimuli and emits a conditioned response to that cue when it is presented. Although it is parsimonious, this system can account for only a few phenomena involving single cues, including acquisition curves, extinction curves, the effect of cue or outcome stimulus salience on the rate of acquisition and extinction (Kamin, 1965), and partial-reinforcement acquisition deficits (Pavlov, 1927). It does not account for cue interaction.

To explain cue competition, Miller and Matzel (1988) proposed that animals use response rules more complex than Equation 2. Specifically, the comparator hypothesis proposes that associatively activated cue competition is mediated at the time of testing by the retrieved memory of competing cues, j , called *comparator stimuli*. Miller and Matzel proposed that the product of the associations between X and j ($V_{X,j}$) and between j and the outcome ($V_{j,O}$) reduce excitation elicited by X 's direct association with the outcome ($V_{X,O}$). These tenets, not originally formalized by Miller and Matzel, are captured in Equation 4. Behavior indicative of inhibition is observed when the product of the X - j and j - O associations is large relative to the strength of the X - O association. The comparison between $V_{X,O}$ and its comparator stimuli is accomplished by subtraction:¹

$$R_X = V_{X,O} - k2 * f(\sum V_{X,j} * V_{j,O}). \quad (4)$$

Equation 4 states that an animal's overt response to a single cue, X , as a predictor of an outcome, will depend on the strength of that cue's direct association with the outcome minus the sum of other associations with that outcome of other cues that are also associated with X . Critical to the retrieval of these competing associations are the associations between X and all cues, j , which act as retrieval bridges through which their associations with the outcome can be activated. The formula is meant to apply only to those trials on which cue X is presented.

Equation 4 states that the response potential of target cue X is proportional to its direct association with an outcome minus the additive contributions of all stimuli j acting as potential comparator stimuli for stimulus X . Because it is the term against which $V_{X,O}$ is compared, the term $\sum V_{X,j} * V_{j,O}$ is called the *comparator term*. The function, f , places an upper limit of +1.0 on the sum of multiple comparator terms, thereby making the positive-valued

¹ We have considered other functions, such as ratios, but subtraction best describes the data that we have simulated. In contrast, Baker, Murphy, Vallee-Tourangeau, and Mehta (2001; also see Murphy et al., 2001) proposed a mathematical implementation of the original comparator hypothesis (Miller & Matzel, 1988), which used a ratio for the critical comparison. This worked with moderate success for them, but they simulated only a small number of phenomena and ignored the role of the within-compound association.

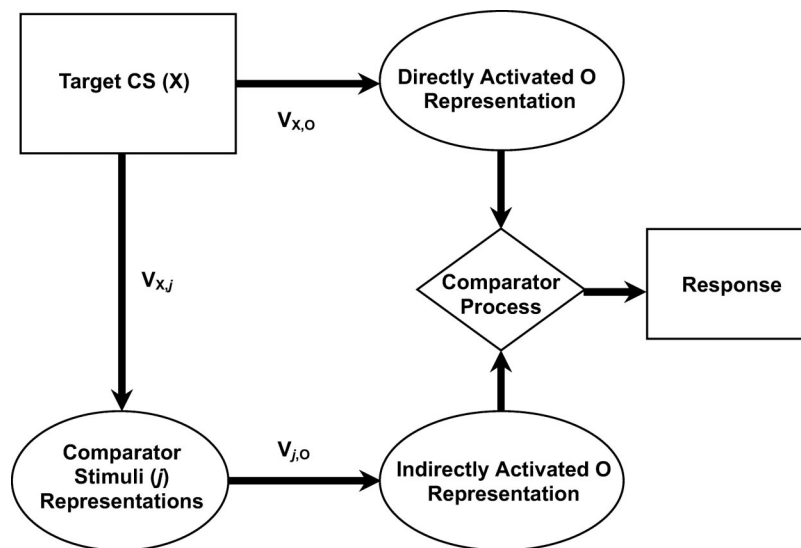


Figure 1. The original comparator hypothesis. Arrows represent the target-cue–outcome association ($V_{X,O}$), the target-cue–comparator-stimulus within-compound association ($V_{X,J}$), and the comparator-stimulus–outcome association ($V_{J,O}$). At test, presentation of the target cue evokes two representations of the outcome, one directly ($V_{X,O}$) and another indirectly mediated through a representation of the comparator stimulus ($V_{X,J}$ and $V_{J,O}$). The strengths of the directly and indirectly activated outcome representations are compared to determine the strength of responding to the target cue. Ovals depict stimulus representations, rectangles depict physical events, and diamonds represent the comparator process.

parameter k_2 the maximum proportion by which responding can be reduced by the comparison process (specifically, the output of f is equal to the parenthetical value if the parenthetical value is less than 1.0, and it is equal to 1.0 if the parenthetical value is equal to or greater than 1.0). These limitations on the size of the comparator effect by f and k_2 account for the observation that cue competition is rarely total and prevent the incorrect prediction that multiple independent competing cues would cause a reinforced cue (X) to become an effective inhibitor (as with A–outcome, B–outcome, followed by ABX–outcome trials; Schachtman, Kaspro, Chee, & Miller, 1985; or with AX–outcome, BX–outcome, CX–outcome trials; Murphy, Baker, & Fouquet, 2001). According to Equation 4, whenever X is presented and $R_X > 0$, excitatory conditioned responding is observed (for purposes of simplification, we here ignore the construct of response thresholds; e.g., Spence, 1938), and whenever $R_X < 0$, behavior indicative of conditioned inhibition can be expected.² The response rule is restricted to the response potential of a single cue. As with most formal models of acquired behavior, we do not supply a precise summation rule for determining responding when multiple cues are presented together (see Wagner, 2003, for a current analysis of summation rules). The associations between cues and the comparison process are represented graphically in Figure 1. Although it is more complex than Equation 2, Equation 4 is a well-behaved performance rule that, when combined with Equation 3, can account for a large number of learning phenomena of a competitive nature. These include overshadowing, blocking (forward and backward), the degraded contingency effect, conditioned inhibition (Pavlovian, explicitly unpaired, and differential), the trial-spacing effect (by equating time in the context between trials with context-alone trials), latent inhibition, the US-preexposure effect, the observation that latent

inhibition and the US-preexposure effect requires that cue preexposure and US preexposure occur in the same context as subsequent conditioning, the overexpectation effect and other competition effects, which have been outlined in numerous papers from this and other laboratories.

Early WYSIWYH models, such as R-W, postulated that modification of a target–outcome association could not occur in the absence of a presentation of the target cue, and the present implementation of the comparator hypothesis shares that premise. However, unlike those early WYSIWYH models, the comparator hypothesis can account for the reversals of these competition effects that are observed when the competing stimulus is extinguished following completion of target training. This family of phenomena are called *deflation effects*, and instances include recovery from overshadowing (e.g., Kaufman & Bolles, 1981), blocking (Blaisdell et al., 1999), conditioned inhibition (e.g., Lysle & Fowler, 1985), the relative validity effect (Cole et al., 1995), and the reversal of the latent inhibition deficit with posttraining extinction of the treatment context (e.g., Grahame, Barnett, Gunther, & Miller, 1994). Finally, the model accounts for phenomena in which the conditioned response to a target CS is weakened when its competing stimulus, punctate or contextual, is further reinforced following completion of target training. Called *inflation effects*, these phenomena occur (e.g., Amundson et al., 2005; Miller & Matute,

² More technically, tests for inhibition are procedurally different and less direct than the simple tests for excitation, which unambiguously reflect R_X , and therefore allow for the possibility that a cue can pass tests for inhibition and still act as a weak excitor, as demonstrated by Matzel, Gladstein, and Miller (1988; for further discussion, see Savastano et al., 1999).

1996; Shanks, 1985) but are decidedly less robust than their deflation-effect counterparts (e.g., Melchers, Lachnit, & Shanks, 2004). The model accounts for this discrepancy in that further reinforcement of $V_{j,O}$ will only decrease R_X provided that $V_{j,O}$ has not been conditioned to asymptote. In contrast, extinction of the $V_{j,O}$ association is always possible. In addition, posttraining extinction of competing stimulus j not only decreases $V_{j,O}$ but also might decrease $V_{X,j}$, whereas posttraining reinforcement of competing stimulus j can increase $V_{j,O}$ but could also decrease $V_{X,j}$. (We later discuss an additional factor that contributes to this asymmetry, which is outside of the framework of the original comparator hypothesis.) In all of the phenomena that this formalization of the comparator hypothesis simulates, the model anticipates modulation by an indirect association between the target cue and the outcome, which can reduce excitatory responding to the target cue and possibly lead to behavior indicative of the target cue being inhibitory, or vice versa. An example of the model's behavior is presented in Figure 2. Depicted is the response potential to the target cue X alone during discrimination training (A–O/XA trials) and during subsequent devaluation of the excitator (A trials). In this simulation, inhibition builds during the discrimination trials (as in R–W) and is abolished by extinction trials (as R–W fails to anticipate). This three-equation formalization of the comparator hypothesis is summarized in Table 1.

Revised WYSIWYH Models

For a few years, the comparator hypothesis was unique in being able to explain the effects of posttraining associative deflation and inflation on responding; thus, the instantiation that we have proposed in Equations 3 and 4 fares well relative to early WYSIWYH models. The original WYSIWYH models failed to account for changes in responding that result from either posttraining deflation or inflation of comparator stimuli (i.e., empirical retrospective revaluation), because they all required that the target cue be present on the trial in question for a change in its associative status to occur. During posttraining revaluation of a competing cue, the

target cue is not present, and the subject can therefore learn nothing new about this cue, according to those models. However, later WYSIWYH models undermined the unique status of the comparator view by adding the assumption that retrospective learning about absent stimuli (i.e., retrospective revaluation of associative strength) can occur, provided an associate of the absent cue (i.e., a stimulus that was present during target training) is present during the revaluation treatment. For example, Van Hamme and Wasserman (1994) proposed a modification of the R–W model, in which the associability (α) of an absent cue is assigned a negative value, rather than a value of zero. The consequence of this modification is that a cue will be incremented or decremented in associative strength (depending upon the sign of the discrepancy term, $\lambda_O - \sum V_i$, of Equation 1), even when that cue is absent on a given trial. That is, extinction of a competing cue (which normally creates a negative discrepancy term because λ is zero) should increase the associative strength of the absent target cue, whereas posttraining reinforcement of a competing cue (which normally creates a positive discrepancy term) should decrease the associative strength of the target cue.

Dickinson and Burke (1996) developed another acquisition-focused account of retrospective revaluation. They revised Wagner's (1981) SOP model by allowing increases in the associative strength between two simultaneously activated stimulus representations, not only when the representations of the two stimuli are activated by the physical presentation of the stimuli (as in the original SOP model), but also when the representations are both activated by presentation of an associate (or associates) of the two stimuli. This modification of SOP accounts for the increase in responding to a target cue because of posttraining deflation of its competing cue, because during deflation treatment, both the target cue and the outcome are absent, but their representations are presumably conjointly activated through retrieval processes. Moreover, the Dickinson and Burke model can account for the decrease in responding to a target cue that results from posttraining inflation of its competing cue, in which the outcome is present and the target cue is absent but reactivated. They argue that an inhibitory association is strengthened when one associate is present, whereas the other is absent, but its representation is activated through retrieval processes. Thus, Van Hamme and Wasserman (1994) and Dickinson and Burke both provide acquisition-focused accounts of cue competition that explain the effects of posttraining associative inflation and deflation, which were previously uniquely predicted by Miller and Matzel's (1988) comparator hypothesis. These new acquisition-focused versions of empirical retrospective revaluation models revitalized the debate over whether cue competition is best viewed as a failure of acquisition or expression (realistically, both are likely involved).

The Problem of Cue Facilitation

A problem shared by the comparator hypothesis and all WYSIWYH models to date, including revised WYSIWYH models, is the old observation of cue-facilitation phenomena, such as second-order conditioning and sensory preconditioning, in which a cue (X) is only paired indirectly with an outcome through another cue (A) and yet is observed to elicit conditioned responding. The reason that models of cue competition do not predict cue facilitation is simple; the mechanisms that account for cue competition tend to

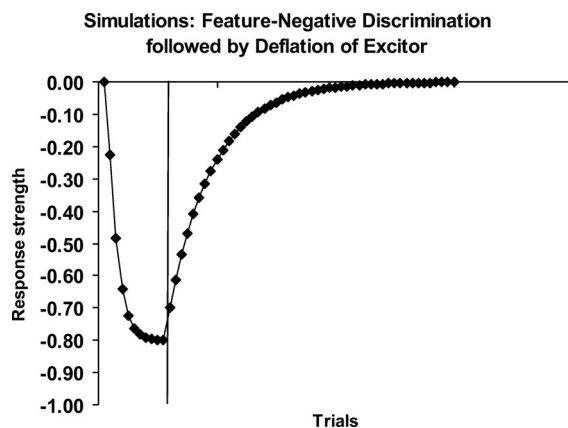


Figure 2. Simulation of the formalized original comparator hypothesis. Depicted is the response potential to the target cue alone during feature-negative discrimination training and during subsequent devaluation of the excitator. Here, stimuli saliences were .71 when presented and 0 when omitted; $k_1 = .10$, and $k_2 = .90$.

Table 1
Equation Systems

Model	Equations
Rescorla–Wagner (Rescorla & Wagner, 1972)	$\Delta V_{X,O} = \alpha_X * \beta_O (\lambda_O \Sigma - V_i)$ $\beta_O = \beta_{up} \text{ when } \lambda_O > 0$ $\beta_O = \beta_{down} \text{ when } \lambda_O = 0$ $R_X \approx V_{X,O}$
Original comparator hypothesis (Miller & Matzel, 1988)	$\Delta V_{S1,S2} = s1 * s2 * (1.0 - V_{S1,S2}), \text{ when } s1 > 0 \text{ and } s2 > 0$ $\Delta V_{S1,S2} = -k1 * s1 * V_{S1,S2}, \text{ when } s1 > 0 \text{ and } s2 = 0$ $R_X = V_{X,O} - k2 * f(\Sigma V_{X,j} * V_{j,O})$
SOCR	$\Delta V_{S1,S2} = s1 * s2 * (1.0 - V_{S1,S2}) \text{ when } s1 > 0 \text{ and } s2 > 0$ $\Delta V_{S1,S2} = -k1 * s1 * V_{S1,S2} \text{ when } s1 > 0 \text{ and } s2 = 0$ $\Delta Op_{X,j,O} = 1.0 - Op_{X,j,O}, \text{ when } V_{X,O} > 0$ $\Delta Op_{X,j,O} = X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}), \text{ when } V_{X,O} = 0$ $R_X = V_{X,O} - k2 * f(\Sigma Op_{X,j,O} * r(V_{X,j}) * r(V_{j,O}))$

Note. V = associative strength; α_X and β_O = learning-rate parameters; λ_O = maximum associative strength supportable by a given outcome on a single trial; R = response; $S1$ and $S2 = s1$ and $s2$ = saliences of $S1$ and $S2$, respectively; $k1$ = a decremental learning-rate parameter ($0 \leq k1 \leq 1.0$) that applies on trials on which the outcome $S2$ is absent; j = all cues except X and O ; $Op_{X,j,O}$ = the switching operator; SOCR = sometimes-competing retrieval.

preclude cue facilitation. The traditional view of cue facilitation is that at test cue X indirectly retrieves a memorial or behavioral representation of A , which in turn is directly associated with O (Brogden, 1939). Evidence that excitatory responding to cue X at test is mediated by the representational chain of X – A –outcome has been provided in experiments in which nonreinforced presentations of cue A have led to reduced responding to X (e.g., Chang, Blaisdell, & Miller, 2003; Rashotte, Griffin, & Sisk, 1977; Rescorla, 1979, 1982; Robertson, Krane, & Garrud, 1984; cf. Holland & Rescorla, 1975; Rizley & Rescorla, 1972). WYSIWYH models do not incorporate this notion of indirect retrieval (but see Wagner, 1981). Although the comparator hypothesis does incorporate this notion, it proposes that indirect retrieval of an outcome representation *attenuates* conditioned responding to the target cue or leads to conditioned inhibition, and this interpretation is also supported by considerable evidence that is detailed above. These models also cannot account for cue-facilitation counterparts of other cue-competition phenomena, such as augmentation (facilitated responding resulting from blocking treatment; e.g., Batson & Batsell, 2000) and potentiation (facilitated responding resulting from overshadowing treatment; e.g., Rusiniak, Hankins, Garcia, & Brett, 1979), or the response tendency controlled by a target cue to change in the same direction as posttraining associative revaluation of the facilitating cue in these paradigms (i.e., posttraining revaluation of a mediating stimulus; e.g., Rescorla, 1982). If the comparator hypothesis's notion of indirect retrieval is to explain cue facilitation, it needs a principled mechanism that allows the comparator term to play either role, competitive or facilitative. In Equation 5, Equation 4 is expanded to include a term called the *switching operator*, or $Op_{X,j,O}$, which determines whether the comparator term is facilitative or competitive:

$$R_X = V_{X,O} - k2 * f(\Sigma Op_{X,j,O} * V_{X,j} * V_{j,O}). \quad (5)$$

$Op_{X,j,O}$ is a variable with a value that can range from -1.0 to 1.0 . Its initial value is -1.0 . Given Equation 5, it can be seen that when

a comparator term's switching operator assumes a negative value, its effect is to facilitate excitatory responding; when the operator assumes a value of 0 , the comparator term has no effect on responding; and when the operator assumes a positive value, the comparator term will attenuate excitatory responding or promote behavior indicative of inhibition. Although the operator is not binary in nature, the name *switching operator* was chosen, because this variable effectively determines whether the mathematical operator between the cue–outcome association and the comparator term is additive or subtractive.

The explanatory power of the model is increased by the addition of $Op_{X,j,O}$ only when combined with a principled rule that specifies how each $Op_{X,j,O}$ changes with experience. The positing of the switching operator, as well as the rule that describes changes in its value, are based on adaptive considerations. An animal best promotes survival and reproductive fitness initially by approaching areas both directly and indirectly associated with appetitive stimuli and by fleeing areas both directly and indirectly associated with aversive stimuli. However, with greater experience, the animal is better served by discriminating between cues that directly signal an outcome and cues that only signal other cues that are actually paired with the outcome. Thus, in situations in which a target cue has been only indirectly associated with the outcome for multiple trials, a switchover to responses to the target cue that are consistent with a decrease in the likelihood of the outcome becomes beneficial. For multiple cues directly paired in compound with the outcome, the same discrimination between direct and indirect associative pathways develops. Here the subject should discriminate among the different individual predictors, because it will have more quickly learned to approach the associated appetitive stimulus or avoid the aversive stimulus on the basis of the better predictors. We can think of two ways of implementing this switch in indirectly activated representations of an outcome, from facilitation to competition, within the context of the comparator hypothesis. One way is through use of the classic notion of surprise (e.g.,

Amsel, 1958; Kamin, 1968, 1969; R-W). During reinforced and nonreinforced compound trials, individual operators for each cue could be adjusted until the discrepancy between the outcome expected on the basis of the direct and indirect associations of all cues presented on that trial and the outcome actually occurring approaches zero. Put more directly, an error-reduction rule similar to R-W could adjust the values of each individual Op. A second possibility is that mere practice with retrieval of the outcome memory forces the conversion from facilitation to competition. In the interest of simplicity, we have tentatively chosen the latter strategy.

Thus, we assume that over successive trials, subjects learn to use directly and indirectly activated representations of the outcome differently. Early, the two types of outcome representations summate, and consequently, the indirectly activated representation facilitates behavior supported by the directly activated representation. Later, the two representations are contrasted, and consequently, the indirectly activated representation attenuates behavior supported by the directly activated representation. This changeover is formally captured by postulating that $Op_{X,j,O}$ has a default (starting) value of -1.0 and increases toward an asymptotic value of $+1.0$ with each successive retrieval of the outcome, on the basis of presentation of the target cue X. As the value of $Op_{X,j,O}$ increases, the effect of the comparator term on the response potential of stimulus X changes from facilitative to competitive/inhibitory. The function relating changes in $Op_{X,j,O}$ to presentations of cue X and to its direct and indirect associations with the outcome is represented as a composite function in Equations 6A (when X and the outcome have ever been paired) and 6B (when X and the outcome have not been paired):

$$\Delta Op_{X,j,O} = 1.0 - Op_{X,j,O}, \text{ when } V_{X,O} > 0, \quad (6A)$$

$$\Delta Op_{X,j,O} = X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}),$$

when $V_{X,O} = 0$. (6B)

Collectively, Equations 6A and 6B state that upon presentation of cue X, two factors have the potential to drive $Op_{X,j,O}$ toward its limit of $+1.0$. One term $V_{X,j} * V_{j,O}$ (Equation 6B), represents X's indirect activation of the outcome representation through all comparator stimuli j and is governed by fractional parameter $k3$, which is set relatively low to reflect indirectly activated representations of the outcome being less effective than directly activated representations of the outcome. Equation 6A applies when X directly activates the outcome representation. Whenever $V_{X,O}$ is nonzero, any deviation between $Op_{X,j,O}$ and its asymptote of 1.0 is added to $Op_{X,j,O}$, causing $Op_{X,j,O}$ to equal 1.0 . This rapid increase in $Op_{X,j,O}$ is consistent with the observed prevalence of cue competition instead of cue facilitation, even with small numbers of trials in paradigms in which X is explicitly reinforced (such as blocking treatment; i.e., A–outcome followed by XA–outcome trials, which differs from phasic second-order conditioning only in that both X and A, as opposed to A alone, are reinforced in the second phase of the treatment).³

The effect of Equations 6A and 6B can be appreciated by considering two contrasting scenarios of phasic second-order conditioning: feature-negative discrimination training (XA–nothing then A–outcome) and sensory preconditioning (A–outcome then XA–nothing). The first treatment tends to make the test cue first

excitatory, then inhibitory. The second treatment tends to make the test cue excitatory only. First, consider how trial order would affect responding when the subject is exposed to segregated X–A and A–outcome pairings. Here, $V_{X,O}$ is zero, because X and the outcome are never paired; this situation calls for application of Equation 6B. If all X–A pairings precede all A–outcome pairings, as in sensory preconditioning, then $Op_{X,j,O}$ (here, j represents A) would remain at -1.0 , because until the test trial, X never had the opportunity to activate directly or indirectly the representation of the outcome. Consequently, excitatory responding to X would be proportional to $V_{X,A} * V_{A,O}$. In this situation, $Op_{X,j,O}$ could not change prior to testing, because changes occur only when X is presented and either $V_{X,O}$ or the product of $V_{X,A}$ and $V_{A,O}$ is nonzero. During Phase-1 pairings, although X is presented and $V_{X,A}$ is positive, $V_{A,O}$ is zero; during Phase-2 pairings, although $V_{A,O}$ is nonzero, X is not presented. However, reversing the order of training, as in second-order conditioning, so that all A–outcome pairings precede all X–A pairings, does cause term $Op_{X,j,O}$ to change. In this case, during the X presentations of Phase 2, the comparator product is nonzero. As a consequence, term $Op_{X,j,O}$ increases, and with sufficient XA trials, responding to X will be transformed from excitatory to inhibitory (e.g., Rashotte et al., 1977; Stout, Escobar, & Miller, 2004; Yin, Barnett, & Miller, 1994). The same rule applies to interspersed X–A, A–outcome pairings, as in Pavlovian conditioned-inhibition treatment. Simulations of this system capture all the phenomena described by Equations 3 and 4, as well as some basic principles of cue facilitation, including second-order conditioning seen early in discrimination training and sensory preconditioning.

Extended Comparisons

The model that we have proposed is encompassed in the system composed of composite learning Equations 3A and 3B, retrieval Equation 5, and composite operator Equations 6A and 6B. This model accommodates the same direct competition and deflation and inflation effects as the system of Equations 2 and 3, but in addition, it is able to predict the occurrence of second-order conditioning, sensory preconditioning, and the transition from second-order excitation to conditioned inhibition that often results as a function of increased numbers of nonreinforced trials, as well as posttraining increases and decreases in responding to a second-order excitatory target when its associate is reinforced or extinguished, respectively, either in second-order conditioning or sensory preconditioning treatments.

The model constructed so far considers modification of the response potential of a target cue by its comparator stimulus. It

³ The change of the switching operator from facilitation to competition with direct reinforcement could be extended to more than one trial for preparations, such as taste-aversion learning, that superficially seem less susceptible to cue competition than to facilitation effects. In other words, in these preparations, a version of Equation 6B (possibly with a different rate parameter) could be applied to direct reinforcement as well as indirect reinforcement. However, we felt that a one-trial change presented here simplified the model for the purpose of explication. This feature of the model precludes a second-order conditioning interpretation of potentiation (i.e., reversed overshadowing). We offer an alternative view of this phenomenon in our discussion of overshadowing treatment.

does not consider the possibility that the effect of a comparator stimulus might be modified by its own comparator stimuli and their corresponding operator values. This feature was proposed by Denniston et al. (2001). They reviewed several experiments that, when interpreted within the framework of the comparator hypothesis, appeared to demonstrate that comparator stimuli can have their own comparator stimuli. For example, B. A. Williams (1996) demonstrated that a blocking cue can itself be blocked (which he used to interpret blocking as a learning deficit). Similarly, Denniston, Savastano, Blaisdell, and Miller (2003) found that overshadowing a blocking stimulus attenuated blocking. In a further experiment, Denniston et al. (2003) challenged Williams's learning-deficit account by demonstrating that blocking to the target cue was restored by extinction of the overshadowing cue, which was a purely second-order comparator stimulus to the target cue, because it had no direct association to the target cue.

The central premise of the *extended comparator hypothesis* (ECH), proposed by Denniston et al. (2001), is that there is competition among potential comparator stimuli for control over behavior. Specifically, second-order comparator stimuli modulate the effectiveness of first-order comparator stimuli, which in turn modulate responding to the target cue. That is, terms $V_{X,j}$ and $V_{j,O}$ of Equation 4 and Figure 1 can be modulated by their own comparator terms. Applying these modifications to Equation 5 results in Equation 7:

$$R_X = V_{X,O} - k2 * f[\Sigma Op_{X,j,O} * r(V_{X,j}) * r(V_{j,O})]. \quad (7)$$

In Equation 7, the associative strengths of each association in the comparator term have been modified by the function r . Function r is of the same form as function R_X presented in Equation 5, except that it applies to any cue—first-order comparator–stimulus association and any first-order comparator–outcome association. To illustrate the application of Equation 7, consider the overshadowing of blocking preparation of Denniston et al. (2003). Their focal condition was traditional blocking in which the nominal blocking cue (A) was overshadowed by a highly salient stimulus (B) in the first phase of training; that is, $BA \rightarrow O$, followed by $AX \rightarrow O$. We ignore the effect of the conditioning context, because they used highly spaced training trials, which should have minimized associations to the context. The resulting associative structure of this treatment is depicted in Figure 3, and the response equation takes the following form:

$$R_X = V_{X,O} - k2 * f[Op_{X,A,O} * r(V_{X,A}) * r(V_{A,O})]. \quad (8)$$

Notice in Figure 3 that this training protocol establishes no comparator stimulus for the $V_{X,A}$ association, so $r(V_{X,A}) = V_{X,A}$. However, a comparator stimulus is established for $V_{A,O}$, and, consequently, $r(V_{A,O})$ can be meaningfully expanded so that

$$r(V_{A,O}) = V_{A,O} - k2 * f(Op_{A,B,O} * V_{A,B} * V_{B,O}). \quad (9)$$

Thus, the potential of $V_{A,O}$ to block responding to X is itself reduced if strong associations are forged between A and B and between B and O. To help the reader appreciate the calculations, in Table 2 we have listed values of each association at the end of three phases of treatment: $BA \rightarrow O$, followed by $AX \rightarrow O$, followed by B alone. If these associations are extinguished by subsequent B-alone presentations, then the value of $r(V_{A,O})$ will approach $V_{A,O}$, and blocking should be observed. Equations of the

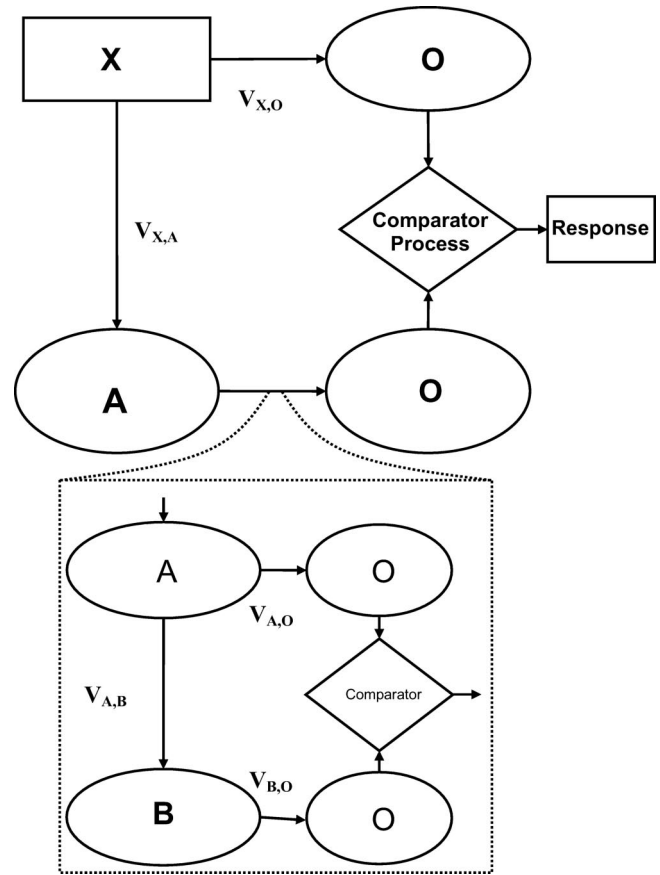


Figure 3. Overshadowing of blocking: Illustration of the extended comparator hypothesis. Associative structure of BA–outcome treatment followed by AX–outcome treatment (with B being higher salience than A or X). X = target cue; O = outcome; V = associative strength; A = first-order comparator stimulus; B = second-order comparator stimulus.

same form as Equation 9 apply to situations that effectively modify $r(V_{X,A})$.

Another thing to notice in this illustration is that the relative impacts of the weighting parameters are not on the same scale. In Equation 8, $k2$ is the weight of all comparator processes, first- and second-order, combined. In Equation 9, $k2$ is the proportional impact of second-order processes relative to first-order comparator processes. Empirical evidence from a variety of studies interpreted in the comparator framework indicates that first-order comparisons can nearly eradicate responding to a conditioned stimulus, and second-order comparisons can nearly eradicate first-order comparisons (i.e., restore conditioned responding to levels similar to those observed with no first-order comparator stimulus), indicating a $k2$ value close to 1.0, although conceptually, the value of $k2$ must be less than 1.0 to damp out the effects of higher-order comparisons.

Equation 9 implies that posttraining associative revaluation of first-order and second-order comparator stimuli should have diametrically opposed effects on responding to the target cue. Consequently, posttraining associative deflation of first-order comparator stimuli should *facilitate* responding to the target cue, whereas posttraining deflation of second-order comparator stimuli should *attenuate* responding to the target cue. Conversely, posttraining

inflation of first-order comparator stimuli should *attenuate* responding to the target cue, whereas posttraining inflation of a second-order comparator stimulus should *facilitate* responding to the target cue. Notice that these two treatments of a second-order comparator stimulus correspond operationally to what have been termed mediated extinction and mediated acquisition, respectively. Relevant to these predictions, De Houwer and Beckers (2002) have examined empirical retrospective revaluation past the second order. In an elegantly designed study in which human participants performed a causal-attribution task, after target training, the researchers independently manipulated first-, second-, and third-order comparator stimuli. Consistent with Denniston et al.'s (2001) version of the comparator hypothesis and the present formalization of it, De Houwer and Beckers found changes in attribution to the target stimulus to be inversely related to the first- and third-order comparator stimuli and directly related to the second-order comparator stimulus. To our knowledge, this is the only report of the effects of systematic manipulation of a third-order comparator stimulus.

Simulations of the Formal Model

We now turn to simulations of the full model. The formal model simulated here includes learning Equations 3A and 3B, the operator Equations 6A and 6B, and the retrieval rule of Equation 7, which are summarized at the bottom of Table 1. The model is named *sometimes-competing retrieval* (SOCR), because sometimes competition is anticipated, and sometimes facilitation (including mediated extinction and acquisition) is anticipated. SOCR belongs to the class of models, such as R-W, that make ordinal predictions, as opposed to the scalar predictions claimed by some models, such as RET or SET. There are two reasons for this limitation. First, the parametric studies necessary to obtain detailed acquisition functions for the compound conditioning situations with which SOCR is centrally concerned are not available; this precludes precise estimates of model parameters. Second, the model is meant to apply to diverse conditioning paradigms, which differ in scales of responding and rates of change. Table 3 summarizes the nature of the different parameters. In the simulations that follow, the following parameter values were consistently used: $k1 = .1768$, $k2 = .9$, $k3 = .1414$. To model trial spacing, we followed a common strategy of representing the duration of the intertrial interval with the number of context-alone trials. Thus, longer intertrial intervals corresponded to more context-alone trials. Similarly, long cue duration was modeled by adding additional trials with that cue. A closer approximation of real-time modeling would require additional equations governing how stim-

Table 3
SOCR Parameters

Parameter	Definition
$k1$	Extinction rate of associations
$k2$	Proportional impact of comparator processes
$k3$	Rate at which operator switches from facilitative to competitive

Note. Although the data reviewed in the present article do not speak to the question, because all the experiments were forms of conditioned suppression, $k1$, $k2$, and $k3$ are likely task dependent. In addition, the saliences of each stimulus (e.g., $s1$ and $s2$) constitute stimulus-specific parameters. SOCR = sometimes-competing retrieval.

ulus trace representations change dynamically with time and experience, a strategy that we here forgo in the interest of parsimony.⁴ Cue saliences were set at 0.0 when a cue was absent and .7071 when present, except for overshadowed cues, which were .3536 when present. The salience of the context was set at .3536. If not stated otherwise in the text or figure captions, trial spacing was an intertrial interval five times the typical cue duration. Other details of simulations are described in the text and figure captions.⁵

Acquisition/Extinction

Figure 4 depicts acquisition and extinction of a conditioned response to a simple cue as simulated by SOCR. A unique property of the acquisition portion of the graph is the preasymptote peak. This phenomenon is commonly observed during Pavlovian conditioning and can be of substantial magnitude (e.g., Bouton, Frohardt, Sunsay, Waddell, & Morris, 2006; Kimmel & Burns, 1975; Miller, Greco, & Vigorito, 1981; Pavlov, 1927; Sherman & Maier, 1978) but has rarely been modeled. According to SOCR, it occurs because the growth in the comparator term lags behind the growth in the X–outcome association itself because it is the product of two fractional associations. Thus, responding reaches a peak prior to a lower asymptote as a result of the delayed growth in the comparator process. This view of the preasymptote peak predicts that extinction of the conditioning context should enhance conditioned responding (e.g., Balsam, 1985), and this effect should be most pronounced following the peak's appearance. Urcelay and Miller (2006b) recently obtained data consistent with this latter prediction. Finally, SOCR predicts that the preasymptote peak should be least pronounced when conditioning takes place in a highly salient context and most pronounced when conditioning takes place in a weakly salient context. The more salient the context, the more growth in the comparator term will approximate growth of the

Table 2
Terminal Associative Strengths Following Described Treatments

Trials	$V_{X,O}$	$V_{X,A}$	$V_{A,O}$	$V_{A,B}$	$V_{B,O}$
100 BA → O	0	0	1.0	1.0	1.0
5 AX → O	.7627	.4871	1.0	.7242	1.0
100 B alone	.7627	.4871	1.0	.7242	0

Note. Parameters used were $k1 = .1768$, $k2 = .9$, and $k3 = .1414$; saliences of cues B and O = .7071; and saliences of cues A and X = .3536. V = associative strength.

⁴ Indeed, expanding SOCR from discrete to continuous time to include representations of stimuli as cascades of distinct stimulus traces, as in the model proposed by Grossberg and Schmajuk (1989), would allow the model to account for many phenomena predicted by the temporal-encoding hypothesis (Savastano & Miller, 1998).

⁵ All programs of the simulations that follow are available at <http://www.jsu.edu/depart/psychology/People/stout/SOCRWEB.htm>. The simulations were written in Microsoft QBASIC 7.1, and the user must have a QuickBasic compiler installed to run the programs.

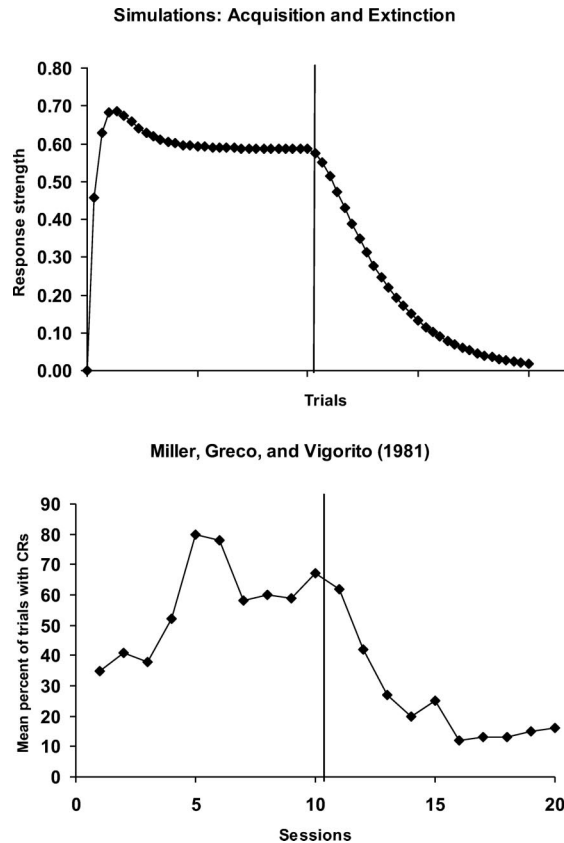


Figure 4. Sometimes-competing-retrieval simulation of simple Pavlovian acquisition and extinction. The preasymptote peak is due to a lag between acquisition of the cue–outcome association and the product of the cue–context and context–outcome associations (i.e., the comparator term). The bottom panel depicts representative rat tail-shock data from Miller et al. (1981). CR = conditioned response.

target cue–outcome association. (Of course, a context of zero salience should exert no competition with the target cue.)

Note that decreases in conditioned responding because of non-reinforced cue exposure following reinforced trials (i.e., extinction) arise from two processes in SOCR. First, Equation 3B posits a decrease in the strength of $V_{X,O}$, and second, cue-training context pairings will strengthen $V_{X,j}$, thereby increasing activation of the indirect representation of the outcome at test. Of course, this latter mechanism will not apply if extinction treatment occurs outside of the conditioning context or if sufficient cue–outcome pairings render $V_{X,j}$ asymptotic.

Overshadowing and Recovery From Overshadowing

As previously stated, the comparator hypothesis explains simple overshadowing as occurring because at test, cue X retrieves the representation of the outcome both directly (through $V_{X,O}$) and indirectly (through $V_{X,j} * V_{j,O}$). The SOCR model explains it in the same manner, with the caveat that the operator term is driven rapidly positive because of the direct retrieval of the outcome by X. However, because it takes one trial for the operator to become positive, SOCR makes the incorrect prediction that one-trial block-

ing or overshadowing should not be observed (cf. James & Wagner, 1980). We consider this prediction to be an artifact of trialwise time steps. The model does, of course, predict the occurrence of simple overshadowing, as well as recovery from overshadowing as a result of extinction of the competing cue (Kaufman & Bolles, 1981; Matzel et al., 1985). SOCR simulations are depicted in the top panel of Figure 5, and data from Matzel et al. (1985) are provided in the bottom panel of Figure 5.

Blocking, Recovery From Blocking, Overshadowing of Blocking, Recovery of Blocking

As previously stated, all contemporary models of learning anticipate blocking (i.e., A–outcome followed by AX–outcome, resulting in attenuated responding to X relative to subjects not receiving the A–outcome pairings). The comparator hypothesis (and SOCR) accounts for blocking in terms of A serving as X’s comparator stimulus. In addition, the comparator hypothesis (and SOCR), as well as select WYSIWYH models of learning (e.g.,

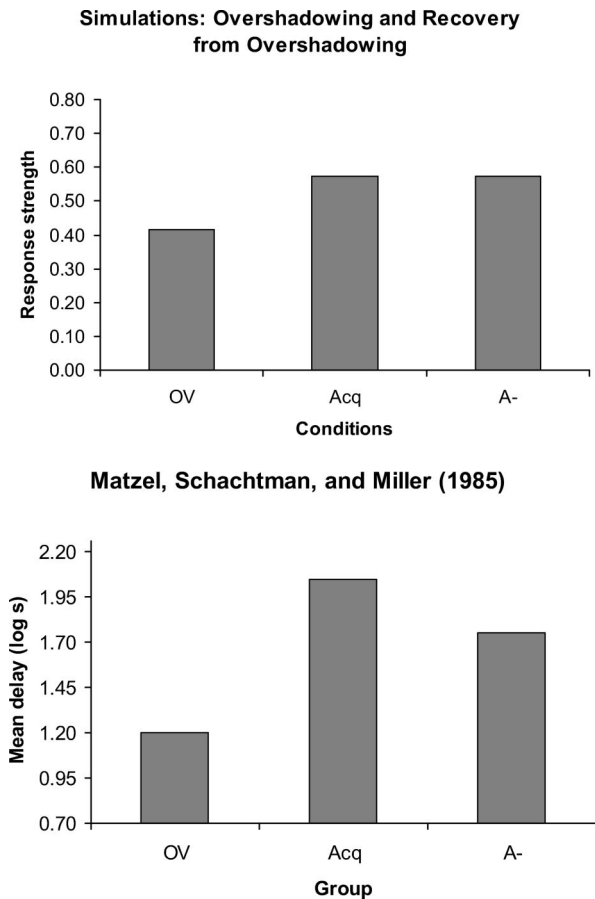


Figure 5. Overshadowing and recovery from overshadowing. The top panel illustrates sometimes-competing-retrieval predictions after five trials. OV represents responding to cue X following overshadowing treatment (AX–O, with A being more salient than X). Acq represents X–O treatment. A– represents the effect of posttraining extinction of cue A on responding to cue X after the same treatment as Group OV. The bottom panel depicts the same conditions from Experiment 1, Matzel et al. (1985).

Dickinson & Burke, 1996; Van Hamme & Wasserman, 1994), correctly predicts that posttraining extinction of A can restore responding to X, as repeatedly reported (e.g., Arcediano et al., 2001; Blaisdell et al., 1999; but see, e.g., Holland, 1999, for evidence that this effect is dependent upon experimental parameters). A prediction of SOCR that differentiates it from the models of Dickinson and Burke (1996) and Van Hamme and Wasserman (1994) is that responding to A is expected to be attenuated by the AX–outcome pairings. Arcediano, Escobar, and Miller (2004) and Hall, Mackintosh, Goodall, and Dal Martello (1977) have confirmed this prediction.

B. A. Williams (1996) found that subjecting a potential blocking cue itself to blocking treatment largely eliminated blocking. A subsequent series of experiments by Denniston et al. (2003) found a similar effect when overshadowing treatment of a cue eliminated its potential to block conditioning of a target cue (with **B** being more salient than A, BA–outcome followed by AX–outcome resulted in robust responding to X). Consistent with the ECH, subsequent extinction of the overshadowing cue (i.e., **B** trials) restored the blocking effect; that is, it led to mediated extinction of responding to cue X. Figure 6 depicts simulations based on SOCR (upper panel) and actual data from Denniston et al. (2003; lower panel).

Cue Preexposure and Overshadowing

Pretraining exposure to a cue is known to retard subsequent acquired behavioral control (i.e., latent inhibition; Lubow & Moore, 1959). In addition, as previously noted, conditioning a target cue in compound with another stimulus is known to retard behavioral control by the target relative to conditioning the target cue alone (i.e., overshadowing; Pavlov, 1927). However, Blaisdell, Bristol, Gunther, and Miller (1998; also see Ishii, 1999) observed facilitation of acquisition to a preexposed cue, if the cue was subsequently trained in compound with a more salient stimulus (X alone followed by AX–outcome). Their experimental parameters were chosen so that, without such preexposure, reinforcement of the target in compound with A led to overshadowing, and without compound conditioning preexposure, it led to a cue-preexposure deficit in responding (i.e., latent inhibition). Alternatively stated, Blaisdell et al. found cue competition deficits normally associated with cue preexposure and overshadowing treatment to counteract each other, rather than to summate, as might be intuitively expected. Figure 7 shows SOCR's simulation of this interaction, with the essential control groups in the upper panel and Blaisdell et al.'s data in the lower panel. According to the ECH/SOCR framework, the two treatments cancel each other, because they allow for the context and overshadowing cue to play roles as first- and second-order comparators simultaneously, with first-order comparisons down-modulating and second-order comparisons up-modulating responding to the target cue. It should be noted that SOCR predicts that only limited combinations of trial spacing and number of training trials will cause this interaction.

Overshadowing Treatment: Paradoxical Interactions With Intertrial Interval and Interstimulus Interval

Two well-known effects in Pavlovian conditioning are that the rate of acquisition of a conditioned response is a direct function of

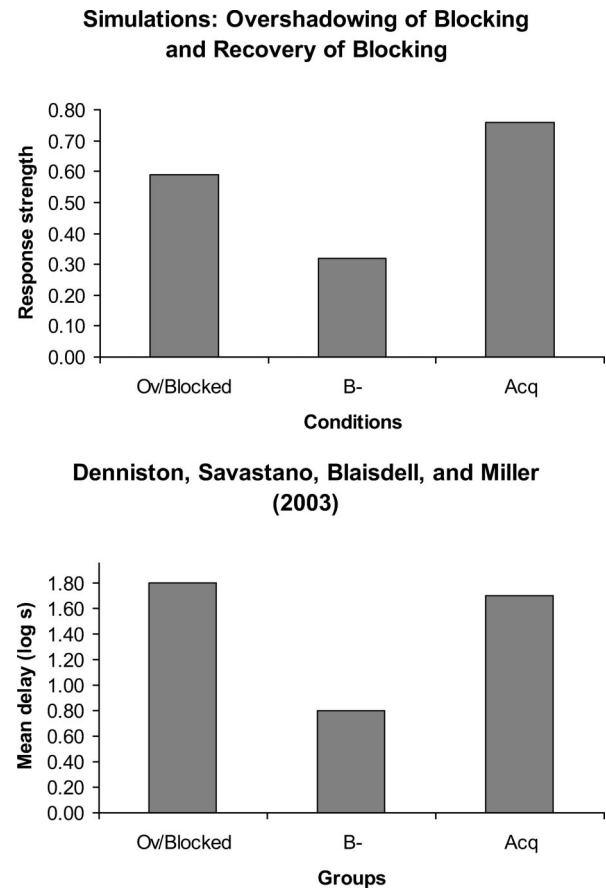


Figure 6. Overshadowing of blocking and recovery of blocking. Top panel depicts sometimes-competing-retrieval simulations after five trials. Ov/Blocked represents BA–outcome treatment followed by AX–outcome treatment (with **B** being more salient than A or X). Acq represents BA–outcome treatment followed by X–outcome treatment. B- represents asymptotic posttraining extinction of cue B after the same treatment as Group Ov/Blocked. Depicted is responding to X alone. The bottom panel depicts conditioned suppression from an empirical test of the same conditions from Experiment 4 of Denniston et al. (2003). Higher suppression times represent more conditioning.

the intertrial interval and that it is an inverse function of the interstimulus interval between cue and outcome onset. In fact, according to several investigators, the tradeoff between these functions is well balanced over a wide range of experimental parameters and procedures, such that the acquisition rate and asymptotic level of Pavlovian conditioning is a function of the ratio of the intertrial interval to the interstimulus interval (reviewed by Gibbon & Balsam, 1981). SOCR does not capture this ratio rule exactly, but it does model the independent effects of intertrial interval and interstimulus interval. Long intertrial intervals allow the context–outcome association (i.e., V_{jO}) to extinguish, thus minimizing the impact of the comparator term, and consequently, a high level of conditioned responding is observed. In contrast, short intertrial intervals prevent extinction of the context–outcome association, increasing the size of the comparator term relative to long intertrial intervals, and therefore, less conditioned responding is observed. Cue duration also manipulates the strength of the comparator term.

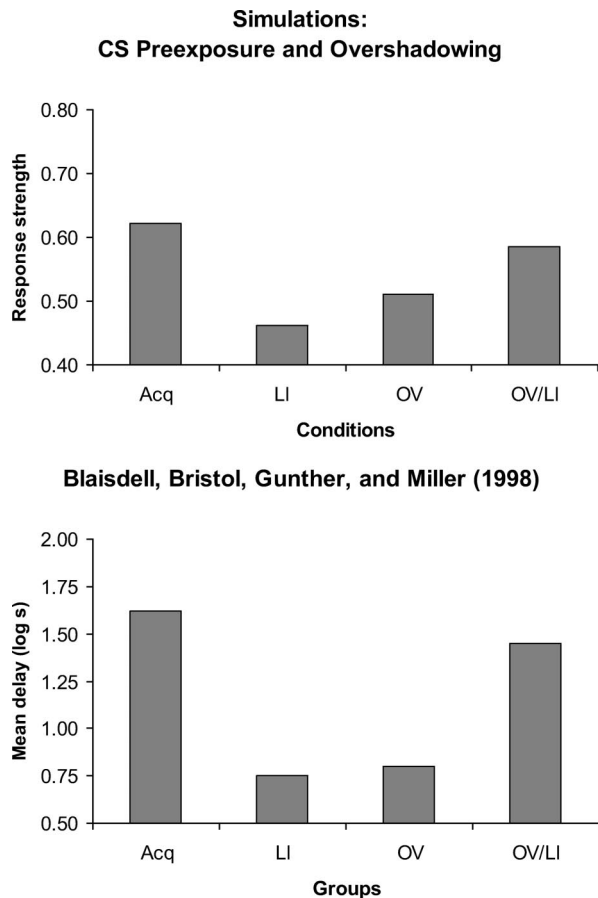


Figure 7. Interactions between LI treatment and OV treatment. Depicted are sometimes-competing-retrieval simulations of responding to cue X following the described training after seven trials. Acq depicts the effect of X–outcome trials. LI represents X trials followed by X–outcome trials. OV represents AX–outcome trials. OV/LI represents X trials followed by AX–outcome trials. The bottom panel depicts the same conditions from Experiment 2 of Blaisdell et al. (1998).

Long-duration target cues have more time to strengthen associations with the context, and consequently, less responding, relative to shorter duration cues, is observed. A set of novel predictions can be made by the ECH/SOCR framework by considering the interaction between these temporal intervals and the number of discrete cues present during acquisition.

As previously stated, the ECH/SOCR model accounts for cue interaction (blocking, overshadowing, Pavlovian conditioned inhibition) with a discrete competing stimulus, A, through response down-modulation exerted by X's indirect retrieval of the outcome representation. However, consider in an overshadowing preparation (AX–O) the additional potential effect of the conditioning context on the X–A and A–outcome associations. To the extent that cue X indirectly retrieves the representation of cue A through the context, the effective direct retrieval of cue A will be reduced (i.e., the effective value of $V_{X,A}$ will become $r[V_{X,A}]$ by subtraction of the term $k_2 * V_{X,C} * V_{C,A}$. Likewise, to the extent that cue A indirectly retrieves the representation of the outcome through the context, cue A's effective retrieval of the outcome will also be

reduced. Because the effective values of $V_{X,j}$ and $V_{j,O}$ for both A and the context are reduced, competition exerted by the comparator term will be reduced, and greater responding will be observed. Thus, the ECH/SOCR model predicts that any operation that enhances the context's potential role as a second-order comparator for the X–outcome association will reduce the overshadowing observed. One such manipulation is massing the reinforced trials, which minimizes extinction of associations to the context during the intertrial interval. With massed trials, the context is predicted to become an effective second-order comparator for cue X and consequently attenuate overshadowing. In addition, the model predicts that subsequent posttraining extinction of the conditioning context will retroactively allow overshadowing by A to emerge. The top panels in Figures 8 and 9 illustrate these predictions. Of course, increasing the associative status of the context not only makes the context an effective second-order comparator stimulus that down-modulates the efficacy of the overshadowing cue as a first-order comparator stimulus for the overshadowed cue; it also makes the context a potential first-order comparator stimulus (as in

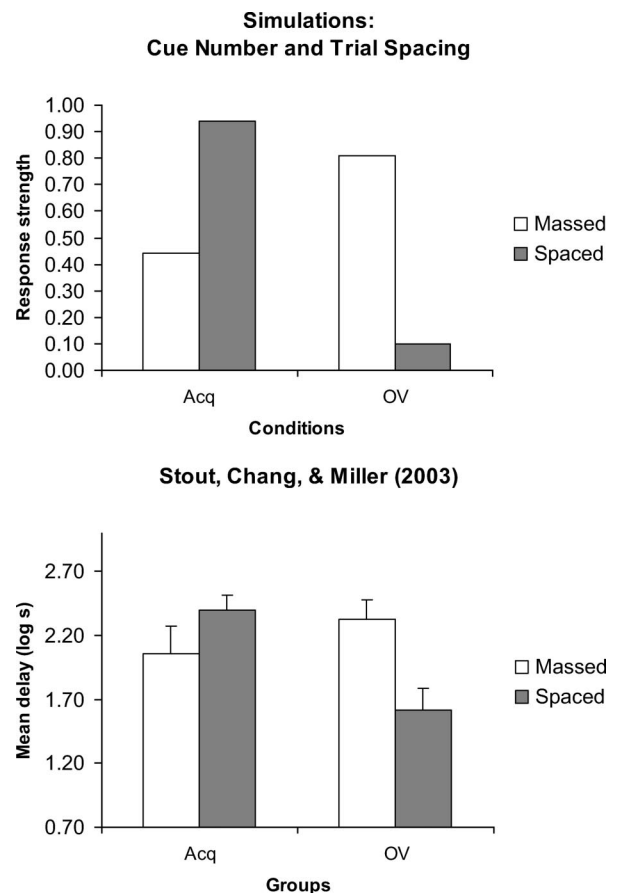


Figure 8. Cue number and trial spacing. Top panel shows sometimes-competing-retrieval simulations of responding to a target cue reinforced elementally (Acq) or in compound with a more salient cue (OV) during massed or spaced acquisition trials. Ten acquisition trials were used. The bottom panel depicts mean conditioned suppression of an empirical test of this prediction by Stout, Chang, and Miller (2003). Error bars denote standard errors of the means.

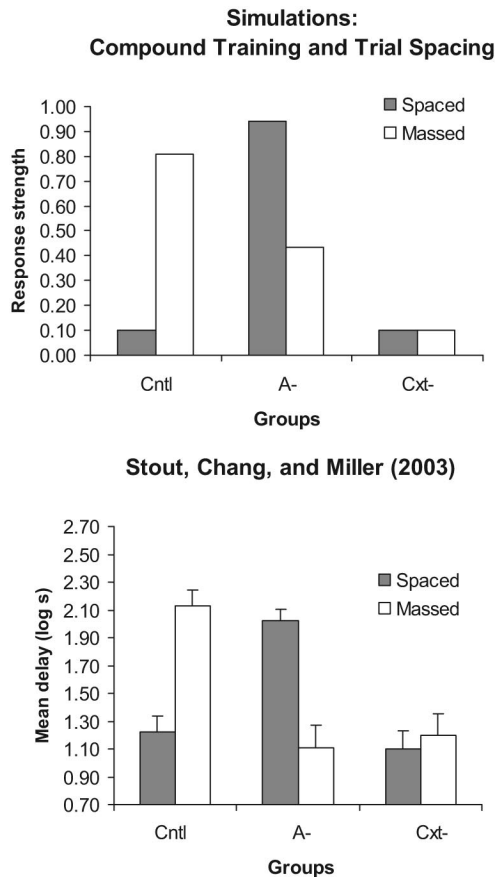


Figure 9. Trial spacing and overshadowing. Top panel shows sometimes-competing-retrieval simulations of responding to a target cue trained in compound with an overshadowing cue (A) during massed versus spaced acquisition trials. Cntl condition received no extinction, A- depicts effect of posttraining extinction of cue A, and Cxt- depicts effect of posttraining extinction of the conditioning context. Ten acquisition trials were used. The bottom panel depicts mean conditioned suppression in an empirical test of this prediction by Stout, Chang, and Miller (2001). Error bars denote standard errors of the means.

simple conditioning). But its efficacy as a first-order comparator stimulus should be down-modulated by the overshadowing stimulus.

Stout, Chang, and Miller (2003) varied trial spacing during overshadowing treatment (i.e., AX–outcome with cue A more salient than cue X) and an elemental control treatment (i.e., X–outcome). They replicated (a) the trial-spacing effect in the elemental control group, in that massed trials impaired conditioned responding relative to spaced trials, and (b) the basic overshadowing deficit in conditioned responding, in that there was less responding to cue X after AX–outcome treatment than after X–outcome treatment. However, counterintuitively but as predicted by ECH/SOCR, not only did the two deficits not summate in the group that received massed overshadowing trials, but the two treatments also counteracted each other (see the bottom panel of Figure 8). That is, whereas the typical overshadowing deficit was observed with spaced trials, a (nonsignificant) tendency toward potentiation (higher responding to the cue reinforced in

compound than to the cue reinforced elementally) was observed with massed trials. As with cue preexposure and overshadowing, apparently massing the trials made the context an effective second-order comparator stimulus that down-modulated the first-order comparator value of the overshadowing cue, while at the same time, the overshadowing cue down-modulated the potential first-order comparator value of the context. To further assess the ECH/SOCR account of the interaction of trial massing and overshadowing treatment, Stout et al. examined the consequences of posttraining associative deflation separately for the context and for cue A. When either of these associative-deflation treatments followed massed overshadowing treatment (i.e., extinction of A or the training context), there was a decrease in conditioned responding to cue X (see the bottom panel of Figure 9). Presumably, deflation of the context permitted cue A to achieve its potential as a first-order comparator stimulus and overshadow cue X; and deflation of cue A permitted the context to achieve its potential as a first-order comparator stimulus and induce a massed trial deficit in responding. These last results give credence to the ECH account of how the two debilitating treatments of trial massing and overshadowing counteract each other.

Urushihara, Stout, and Miller (2004) conducted a similar experiment examining the effect of cue duration on acquisition of conditioned responding to a cue trained in compound, as opposed to elementally. In SOCR, increased duration of overlap of the target stimulus, overshadowing stimulus, and context increases the potential of each to attenuate the response-reducing potential of each other through the same mechanisms as massed trials. That is, as second-order comparator stimuli, both the context and overshadowing cues are each better able to retrieve a representation of the other stimulus, thereby reducing each cue's potential to serve as a first-order comparator cue. Therefore, SOCR predicts a rough functional equivalence between a long-duration-cue-and massed-cue–outcome trials. (This rough equivalence between interstimulus interval and intertrial interval breaks down in simulations involving exceedingly long cues because of extinction of the target cue–outcome association.) Thus, the model predicts that long cue duration and overshadowing treatments should counteract each other (see Figure 10, top panel). Using a conditioned lick suppression preparation with rats, Urushihara, Stout, and Miller found this interaction (see Figure 10, bottom panel). In other words, Urushihara, Stout, and Miller observed potentiation, rather than the usual overshadowing. Taken together, these results suggest that the well-known control exerted by the ratio of intertrial interval to target-cue duration (the I/T ratio; Gallistel & Gibbon, 2000; Gibbon & Balsam, 1981) actually reverses when cues are trained as members of a stimulus compound. Whether responding to cues reinforced in compound scales as directly to the T/I ratio over a broad range of time values as responding to cues reinforced elementally scales directly to the I/T ratio remains to be determined. Also not yet determined is whether potentiation in general (e.g., Rusiniak et al., 1979) depends upon competition between the overshadowing cue and the conditioning context at the time of testing.

Cue Competition and Extended Training

In a series of conditioned-suppression studies with rats, Stout, Arcediano, Escobar, and Miller (2003) considered the conse-

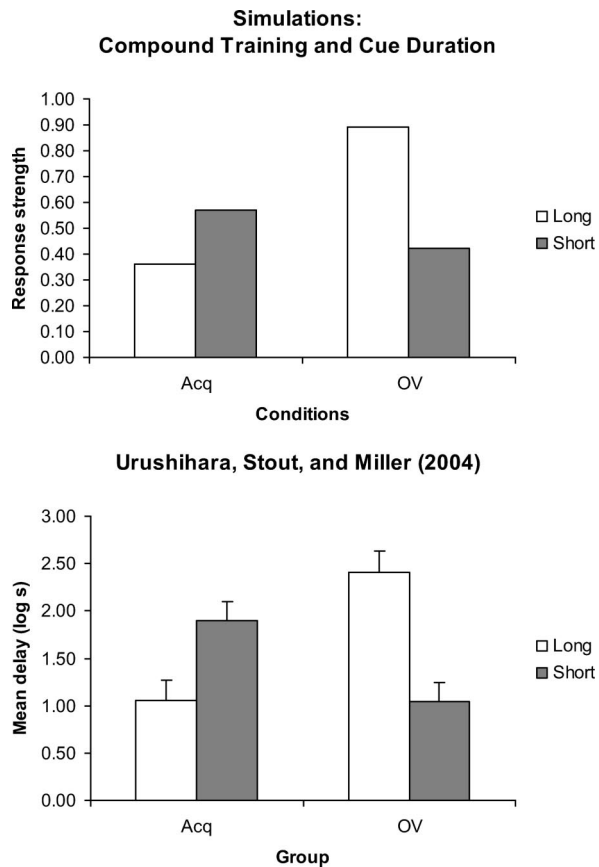


Figure 10. Trial spacing and cue duration. Top panel shows simulated conditioned responding to a target cue (X) trained elementally (Acq) or in compound (OV) with a more salient overshadowing cue (A) during reinforced trials. Cues were long (3 time units) or short (1 time unit) in duration. The bottom panel depicts an empirical test of this prediction by Urushihara, Stout, and Miller (2004). The ordinate represents time to return to drinking in the presence of cue X. Higher scores represent more conditioned excitation. The duration of the cues in the conditioning phase was 25 s in Groups OV-long and Acq-long, and was 2.5 s in Groups OV-short and Acq-short. Error bars denote standard errors of the means.

quences of two assertions made by Miller and Matzel (1988) concerning the comparator hypothesis. First, the model assumes that associative strengths are fractional and grow according to a negatively accelerated function because of the incremental learning process represented in Equation 3A. Second, comparator terms for a given association are always products of at least two associative strengths. These considerations mean that first-order comparator effects grow according to the products of two negatively accelerated fractional functions, constituting a sigmoid shape. Likewise, second-order comparator effects are the products of four such functions, constituting an elongated sigmoid. These relationships anticipate that first-order comparator effects should lag in time behind the growth of the target CS–outcome association, as occurs in the preasymptote peak discussed previously. Similarly, second-order comparator effects should take more training trials than first-order comparator effects to become evident. Thus, with sufficient training, second-order comparator effects of the training

context should become more pronounced, thereby diminishing first-order comparator effects of a punctate cue that is a companion of the target cue during training. Because SOCR is implemented with the aforementioned two assumptions, it makes the same prediction. This is demonstrated in the simulations depicted in the top panel of Figure 11, with overshadowing as the treatment. To test this prediction, Stout, Arcediano, et al. (2003) gave small and large numbers of overshadowing trials (i.e., AX–outcome) and found that, with sufficient training trials, overshadowing waned (see the bottom panel of Figure 11). In a second experiment, they further demonstrated that posttraining associative deflation of the training context restored overshadowing of cue X, as predicted by the ECH/SOCR framework. To our knowledge, other than the ECH, there is currently no model that can account for these observations.

A closely related and also unique prediction of SOCR is that, after blocking treatment (i.e., A–outcome trials followed by AX–outcome trials), large numbers of additional A–outcome trials will attenuate blocking, that is, enhance responding to the target cue (X). As in the case of overshadowing described in the preceding paragraph, this expectation arises from the additional A–outcome

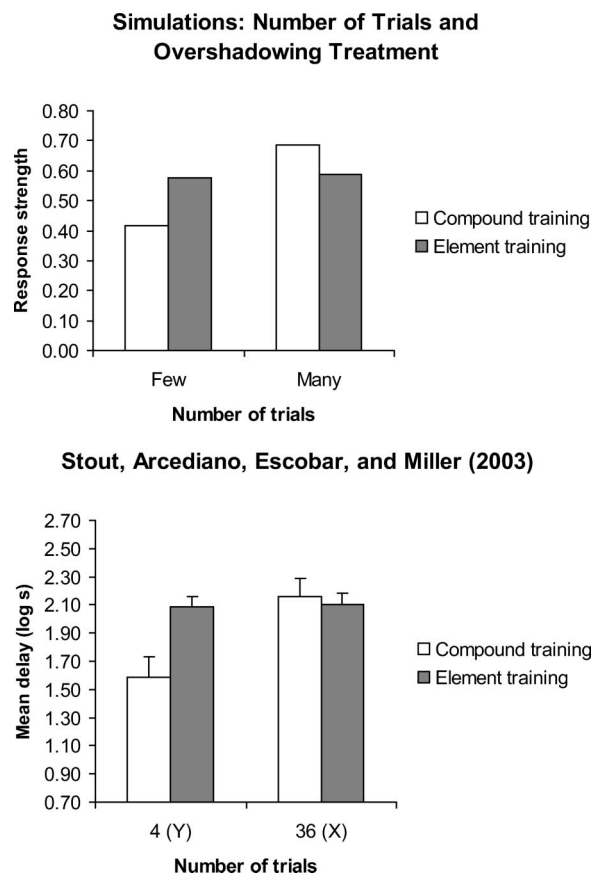


Figure 11. Overshadowing and trial number. The top panel depicts the sometimes-competing-retrieval model's simulation of overshadowing treatment as a function of the number of training trials. Bottom panel depicts the results of a conditioned lick suppression experiment by Stout, Arcediano, Escobar, and Miller (2003). Higher scores indicate more conditioned suppression. Error bars denote standard errors of the means.

pairings, making the context a more effective comparator stimulus for A and, thus, a more effective second-order comparator stimulus for the target cue. The same elongated sigmoid for second-order comparator values that accounts for reduced overshadowing given many overshadowing trials also causes SOCR to predict this waning of blocking with added A–outcome pairings. This prediction is even more counterintuitive than that of the preceding paragraph concerning extended overshadowing treatment, because here, on the added A–outcome trials, the target stimulus (X) is not being presented. Pineño, Urushihara, Stout, Fuss, and Miller (2006) recently confirmed this prediction. In contrast, Dickinson and Burke (1996) and other advocates of acquisition-focused models of retrospective revaluation anticipate that A–outcome trials following blocking treatment will, if anything, further attenuate responding to X. Consistent with SOCR, Pineño, Urushihara, Stout, et al. also demonstrated that posttraining extinction of the training context restores blocking that had been lost as a result of extended A–outcome pairings.

Second-Order Conditioning and Conditioned Inhibition

Since Pavlov (1927), researchers have wrestled with the construct of conditioned inhibition, which in contemporary empirical usage refers to an acquired tendency of a stimulus to attenuate responding to a conditioned excitator with which it is compounded (summation test) and to require more excitatory training trials than an appropriate control stimulus to become a conditioned excitator (retardation test). There have been many attempts to conceptualize conditioned inhibition theoretically. These have included cue–no–outcome associations (e.g., Konorski, 1967), negatively valued cue–outcome associations that coexist with positively valued cue–outcome associations (e.g., Pavlov, 1927; Pearce & Hall, 1980; Wagner, 1981), negatively valued cue–outcome associations that exist on the same dimension with positively valued cue–outcome associations (making excitation and inhibition mutually exclusive; e.g., Rescorla & Wagner, 1972), and inhibitory cues elevating the threshold for activation of the outcome representation required to generate an excitatory response (Konorski, 1948). It is notable that each of these conceptualizations postulates mechanisms above and beyond those required to account for excitation. Moreover, these mechanisms are vague, because there do not seem to be any direct tests of these hypothetical mechanisms underlying conditioned inhibition (Savastano, Cole, Barnet, & Miller, 1999). The comparator hypothesis (original and extended) avoids postulating a special mechanism explicitly for conditioned inhibition. Instead, it treats behavior normally viewed as indicative of conditioned inhibition as responding appropriate for a cue that signals a lower excitatory expectation of the outcome, relative to the excitatory expectation of the outcome on the basis of the other cues that were present during training of the cue. As can be seen in Figure 1, posttraining treatments that increase the associative strengths of these comparator (companion) cues to either the target cue ($V_{X,j}$) or the outcome ($V_{j,O}$) should increase behavior indicative of inhibition, and posttraining treatments that decrease the associative strengths of these comparator cues to either the target cue ($V_{X,j}$) or the outcome ($V_{j,O}$) should decrease behavior indicative of inhibition. Both the former prediction (Amundson et al., 2005; Baker, Murphy, & Mehta, 2003; Chapman, 1991; D. A. Williams & Docking, 1995) and the latter prediction (e.g., Best, Dunn, Batson,

Meachum, & Nash, 1985; Kasproff et al., 1987; Lysle & Fowler, 1985; Miller, Hallam, Hong, & Dufore, 1991; Schachtman, Brown, Gordon, Catterson, & Miller, 1987) have been confirmed. Moreover, presentations of a conditioned inhibitor alone (i.e., operational extinction), following inhibitory training, have been found to increase inhibitory value (e.g., DeVito & Fowler, 1986, 1987; Hallam et al., 1992; Holland & Gory, 1986; Pearce, Nicholas, & Dickinson, 1982; D. A. Williams, Travis, & Overmier, 1986). This is consistent with the comparator hypothesis, at least for some forms of conditioned-inhibition training (e.g., explicitly unpaired and differential inhibition treatments). Hence, the framework of the comparator hypothesis offers a parsimonious account of conditioned inhibition.

The top panel of Figure 12 presents simulations of feature-negative training (i.e., interspersed A–outcome and XA trials). Notice that initially the indirect retrieval of the outcome representation by stimulus X that is mediated through cue A leads to anticipation of second-order excitation, because the initial value of the operator term is negative. With extended practice, the operator becomes positive, and consequently, conditioned inhibition is expected after a point of neutrality is passed. This transition from second-order conditioning to conditioned inhibition is predicted to occur more rapidly with simultaneous XA pairings than when X precedes A, because X is more salient in the former case than in the latter (on the basis of the assumption that a just-presented but now absent cue leaves a decaying stimulus trace of lesser salience than when the cue was presented). Stout et al. (2004) observed this pattern in conditioned-suppression experiments with rats, thereby confirming and extending Yin et al.'s (1994) observation that amount of training is a key variable in determining the transition from second-order conditioning to conditioned inhibition. The middle and bottom panels of Figure 12 depict Stout et al.'s excitatory test data and their retardation test data for conditioned inhibition. Space precludes inclusion of their summation test data, but it was fully consistent with the retardation test.

Another finding consistent with the SOCR interpretation of the transition from second-order conditioning to conditioned inhibition is the effect of proportion of trials in which X and A are jointly presented relative to the total number of A–outcome and XA trials. Rashotte, Marshall, and O'Connell (1981) found conditioned inhibition to emerge more quickly with a higher proportion of XA trials. According to SOCR, precisely this pattern is expected, because it is the XA trials that drive the operator switch to a positive value through indirect retrieval of the outcome representation.

Superconditioning and Overexpectation

One of the major successes of the R-W model was its a priori prediction of the phenomenon of superconditioning. *Superconditioning* refers to enhancement in conditioned responding to a cue (X) as a consequence of that cue having been paired with the outcome in the presence of a conditioned inhibitor (i.e., AX–outcome, with A having been previously trained [and sometimes concurrently maintained] as a conditioned inhibitor; e.g., B–outcome trials interspersed with nonreinforced AB trials), relative to subjects that received X–outcome pairings in the presence of a neutral stimulus C (i.e., CX–outcome). However, Navarro, Hallam, Matzel, and Miller (1989) demonstrated that conditioned

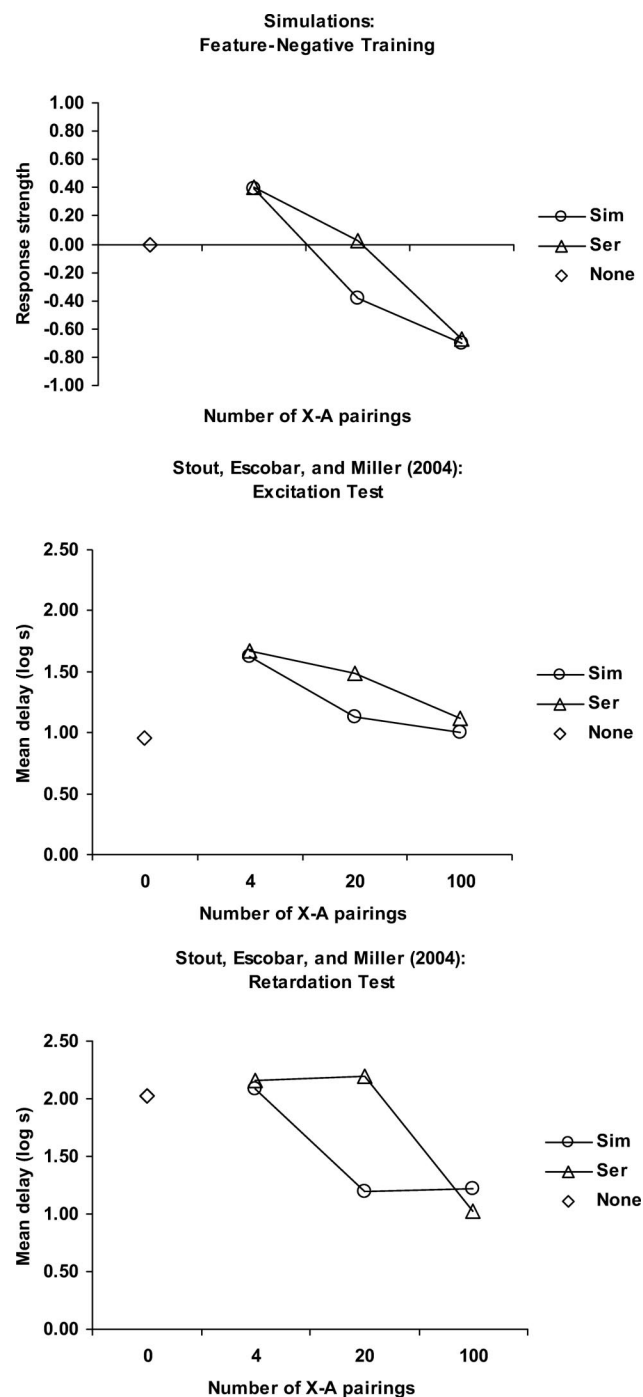


Figure 12. Feature-negative training. The top panel depicts the predictions of the extended comparator hypothesis/sometimes-competing-retrieval model concerning the interaction of cue duration and number of nonreinforced XA-alone trials interspersed among A–outcome (i.e., feature-negative training) trials. The middle panel depicts the results of an excitatory test of this prediction by Stout, Escobar, and Miller (2004). The bottom panel depicts a retardation test for conditioned inhibition with lower scores indicative of greater conditioned inhibition. In both the middle and lower panels, X and A were presented serially (Ser) or simultaneously (Sim) for 0, 4, 20, or 100 pairings.

responding to cue X following AX–outcome pairings was no greater than responding by subjects that merely received X–outcome pairings. Navarro et al. interpreted their results as being consistent with the Pearce and Hall (1980) model by assuming that the prior inhibitory training of cue A resulted in decreased attention to cue A, which in turn reduced the potential of cue A to overshadow cue X during the subsequent AX–outcome pairings. The ECH provides an alternative explanation of the prevention of overshadowing following superconditioning treatment. According to the ECH, the Pavlovian conditioned-inhibition treatment (i.e., B–outcome trials interspersed with nonreinforced AB trials), used to establish cue A as a conditioned inhibitor, produces a strong B–outcome association. This B–outcome association then down-modulates the comparator status of cue A. That is, B serves as cue A’s comparator stimulus (second-order comparator stimulus to cue X) and allows the X–outcome association to be expressed at the time of testing. This view leads to novel predictions. It anticipates that, following superconditioning treatment, extinction of either cue A or cue B should decrease conditioned responding to X (eliminate superconditioning). The top panel of Figure 13 shows SOCR’s simulation of this procedure. Recently, these predictions were confirmed by Urushihara, Wheeler, Pineño, and Miller (2005) in a taste-aversion preparation with rats (see the bottom panel of Figure 13).

Another success of the R-W model was its account of the phenomenon of overexpectation. Overexpectation refers to the decrement in responding to a previously trained cue seen as a result of reinforcing it in compound with another conditioned excitator (i.e., A–outcome interspersed with X outcome, followed by AX–outcome; e.g., Rescorla, 1970). However, subsequent models have also been able to explain it. The comparator hypothesis does so by assuming that during the AX–outcome trials, A becomes a first-order comparator stimulus for X and vice versa. Blaisdell, Denniston, and Miller (2001) demonstrated that, following the AX–outcome trials, the extinction of A attenuated the overexpectation effect, an observation consistent with both the comparator hypothesis and acquisition-focused models of retrospective revaluation (e.g., Dickinson & Burke, 1996; Van Hamme & Wasserman, 1994). Further experiments reported in Blaisdell et al. (2001) pose problems for these acquisition-focused models, but space precludes reviewing them here.

A common assumption since the R-W model was proposed is that when two cues are trained together, despite the two cues having initially different prior associative status because of prior elemental training, they will undergo identical changes in associative status. In a cleverly designed experiment closely related to superconditioning, Rescorla (2000) recently demonstrated that this is not always true. He initially trained one cue as a conditioned excitator and another cue as a conditioned inhibitor. Then he compounded the two cues and either reinforced the compound or did not reinforce the compound. Reinforcement resulted in a greater gain in associative strength by the inhibitor than the excitator, and nonreinforcement resulted in a greater loss in associative strength by the excitator than the inhibitor. Rescorla did not champion any specific account of this remarkable finding, which is contrary to the central premise of the R-W model. He did not consider the possibility that changes in the associative strength of each stimulus element occur independently, presumably because such an assumption

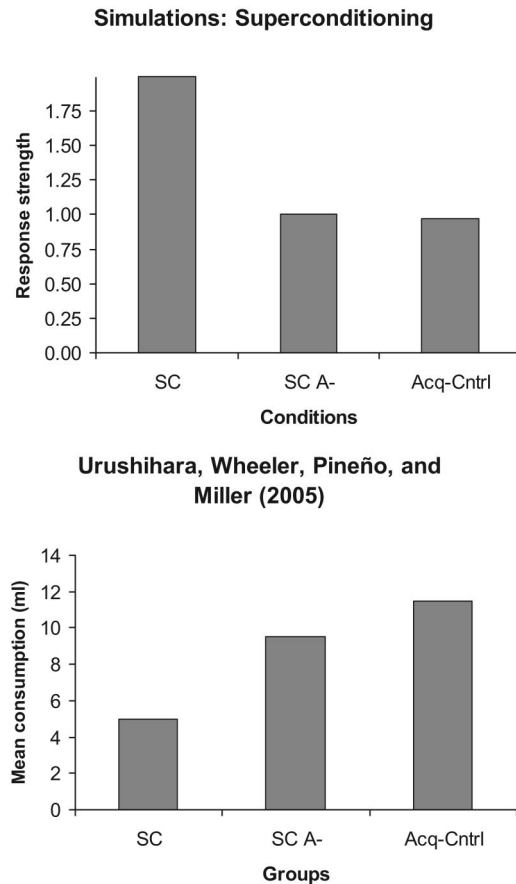


Figure 13. Superconditioning. The top panel depicts sometimes-competing-retrieval simulations. SC denotes superconditioning treatment, that is, conditioning of a compound following asymptotic inhibitory training of the nontarget member of the compound. SC A- represents subsequent extinction of the excitatory stimulus used to establish the inhibitory companion of the target. Acq-Cntrl is an acquisition control. The bottom panel depicts the same data from Experiment 3 of Urushihara, Wheeler, Pineño, and Miller (2005). This experiment used a conditioned taste-aversion paradigm, so here, lower consumption represents more conditioning. Thus, the simulation and data do approximate each other.

would fail to account for simple cue competition, such as overshadowing and blocking, in terms of an acquisition deficit. However, the comparator hypothesis (including SOCR) explains cue competition in terms of a performance deficit; hence, there is no need to cling to an acquisition-failure account of cue competition. Indeed, Equation 2 fully anticipates Rescorla's finding that cues trained together do not necessarily undergo common changes in their associative status. A cue that starts from a lower baseline of associative strength (zero associative strength for an inhibitor in the present framework) should gain more associative strength when it is reinforced than one that starts from a higher baseline (an excitor). Conversely, an excitor that is extinguished should lose more associative strength than should an inhibitor, which should lose no associative strength when extinguished. With many compound trials, predictions would grow more complex as an association developed between the inhibitor and excitor. But with a small number of

compound trials, especially compared with the numbers of prior elemental trials, SOCR fully anticipates Rescorla's results.

Degraded Contingency

An additional finding that is readily accommodated by the ECH/SOCR framework is that the degraded contingency effect (Rescorla, 1968) can be prevented by using a punctate cue (i.e., a cover stimulus) other than the target to signal the added outcomes (e.g., Durlach, 1983; Shanks, 1986, 1989). As previously discussed, the original comparator hypothesis explains the degraded contingency effect as arising from the context serving as the target cue's comparator stimulus and the added unsignaled outcomes of the degraded contingency treatment, making the context more excitatory than in the absence of the added unsignaled outcomes. The effect of signaling the added outcomes with a cover stimulus is readily explained by SOCR, because the cover stimulus serves as a comparator stimulus for the context (i.e., the signal is the second-order comparator stimulus for the target cue), thereby undermining the effectiveness of the context as the first-order comparator stimulus for X. It is notable that most acquisition-focused models can also explain both of these effects in terms of blocking of the cue–outcome association by the context–outcome association and the added cue–outcome association overshadowing the context–outcome association. Thus, here the comparator hypothesis performs as well as, but no better than, acquisition-focused models.

The ECH and WYSIWYH models of learning diverge in their predictions, however, concerning how unsignaled outcomes interspersed among target-cue–outcome pairings (i.e., degraded contingency treatment) should interact with the presence of another more salient cue being compounded with the target cue during training (i.e., overshadowing treatment). Contemporary WYSIWYH models (both conventional and those designed to account for retrospective revaluation) anticipate that the effects of these two response-attenuating procedures should summate, yielding a response deficit greater than either effect alone. In contrast, ECH/SOCR anticipates that the excitatory training context will down-modulate the overshadowing stimulus's potential as a first-order comparator stimulus for the target cue, and the overshadowing stimulus will down-modulate the context's potential as a first-order comparator stimulus for the target cue (see the top panel of Figure 14). The principle here is identical to the previously discussed counteraction between trial massing and overshadowing treatment. Recently, Urcelay and Miller (2006a) obtained data supportive of this prediction with rats in a conditioned lick-suppression preparation. Moreover, consistent with the ECH, they found that posttraining extinction (associative deflation) of the context allowed emergence of an overshadowing effect and that posttraining extinction of the overshadowing stimulus allowed emergence of a degraded contingency effect (see the bottom panel of Figure 14, which represents Urcelay & Miller, 2006a). In addition, SOCR is unique in correctly predicting that posttraining extinction of the signal for the added outcomes will reduce responding to the target cue (Witnauer & Miller, in press). Even the acquisition-focused models designed to account for retrospective revaluation (e.g., Dickinson & Burke, 1996; Van Hamme & Wasserman, 1994) fail to anticipate this, because there is no direct association between the target cue and the signal for the added outcomes.

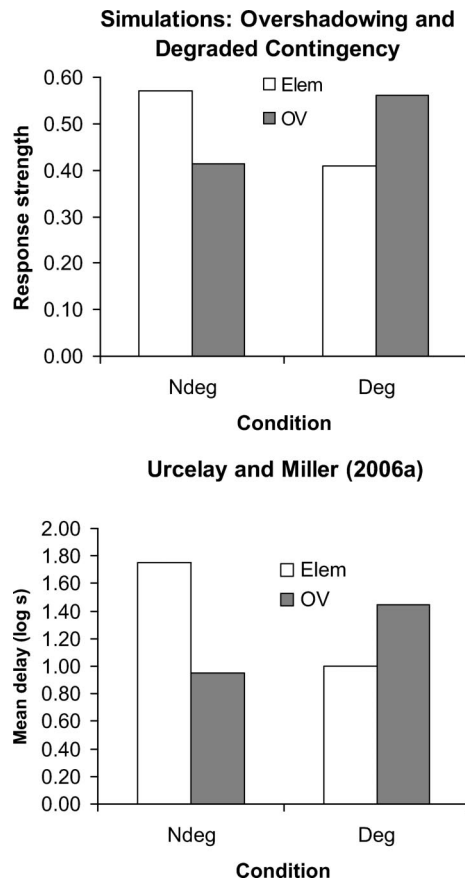


Figure 14. Degraded contingency and overshadowing. Deg represents degraded contingency trials (i.e., trials with unsignaled outcome presentations interspersed among the target-cue–outcome trials); Ndeg represents nondegraded contingency. OV represents overshadowing trials (AX–outcome); Elem represents elemental trials (X–outcome). Top panel depicts simulations of sometimes-competing retrieval using two unsignaled outcome presentations per signaled outcome presentation. The bottom panel depicts Experiment 1 of Urcelay and Miller (2006a).

A challenge to the SOCR account of the cover-stimulus effect is the suggestion that the relative durations of the target cue and cover stimulus determine the magnitude of the effect. Jakubow, Brown, and Hemmes (2004) reported that the effect was maximal when the duration of the cover stimulus was equal to that of the target cue. When it was substantially shorter or longer, degraded contingency was observed. SOCR can only crudely simulate this effect by assuming that, through stimulus adaptation, variation in effective stimulus intensity follows a positively skewed function of time. Thus, exceedingly long or short cover stimuli would have little effective salience, and little or no cover stimulus effect would be observed. However, this idea does not explain why the duration of the most effective cover stimulus scales to the duration of the target cue, as Jakubow et al. observed.

Outcome Competition

Although the comparator hypothesis was originally designed to account for interactions between cues trained together (e.g., over-

shadowing, blocking, and conditioned inhibition), the model has proven flexible by illuminating phenomena for which it was not originally intended. For example, despite the intense focus in recent decades on interactions between cues that were trained together, there have been reports of competition between outcomes presented together. For instance, overshadowing of outcomes is evidenced when cue X in training is paired with two outcomes (O1 and O2 presented simultaneously), and the resultant stimulus control indicative of the X–O1 association is found to be attenuated relative to when cue X is paired with O1 in the absence of O2 (Rescorla, 1980). In addition, blocking of outcomes is evidenced in attenuated stimulus control indicative of an X–O1 association when cue X is initially paired with O2 before cue X is paired with a simultaneous compound of O1 and O2, relative to a group lacking the initial X–O2 pairings (Esmoris-Arranz, Miller, & Matute, 1997; Miller & Matute, 1998). It is notable that this sort of overshadowing and blocking of outcomes is observed, even when measures have been taken to preclude response competition (i.e., the observed competition between outcomes is not simply a matter of a subject not being able to do two things at once) or competition for attention (i.e., the competition is not due to O2 distracting the subject from O1). The comparator hypothesis's account of competition between outcomes hinges on cues and outcomes both being stimuli with neural representations that can enter into associations with any other contiguous stimulus representation. Predictive value (i.e., a cue preceding an outcome) may be necessary for the behavioral expression of most associations, but it is not necessary for the formation of a stimulus–stimulus association (e.g., Savastano & Miller, 1998). Hence, to the extent that one is analyzing a cue X–O1 association, the occurrence of O2 in compound with O1 is treated in the framework of the comparator hypothesis in the same way as if O2 was a second cue. Hence, the O2–O1 association (in conjunction with the X–O2 association) competes with the X–O1 association when cue X is presented at test. Indeed, simulations of reciprocal overshadowing between outcomes demonstrate identical response values to those of overshadowing between predictor cues. Consistent with this comparator view of competition between outcomes, Wheeler and Miller (2005) found that extinction of the competing outcome restored behavior indicative of the X–O1 association.

Figure 15 provides a partial list of observed phenomena that we have simulated with SOCR to date.

Implications

Here we have proposed a quantitative extension of the comparator hypothesis, the goals of which are fourfold: (a) to formalize the ECH; (b) to elaborate the process by which comparator stimuli effectively compete for behavioral control; (c) to generate testable predictions that could be differentiated from those of WYSIWYH models of retrospective revaluation; and (d) to account not only for the cue competition and conditioned inhibition anticipated by the ECH, but also for the cue facilitation that is observed in phenomena such as second-order conditioning. The novel feature of the ECH is the assumption that all nontarget stimuli compete against each other for modulatory control of responding to a target cue. That is, all potential comparator stimuli act both as first-order comparator stimuli, and as second-order comparator stimuli that modulate each other's effectiveness as first-order comparator stim-

Phenomena of simple conditioning:

Acquisition function
Preasymptote peak during acquisition
Enhancement of simple conditioned responding following context deflation
Attenuation of simple conditioned responding following context inflation

Extinction function (including its normally being more gradual than the equivalent acquisition function)
Partial recovery from extinction following context deflation
Trial-spacing effect
Interstimulus-interval effect (excluding the simultaneous-conditioning deficit)

Phenomena related to blocking:

Blocking effect
Inverted-U influence of number of A–O trials given pre- or postcompound trials
Backward blocking is weaker than forward blocking.
Waning of blocking with extended compound trials
Blocking between outcomes

Phenomena related to overshadowing:

More robust with differential stimulus salience
Potentiation instead of overshadowing with long-duration conditioned stimuli
Weakened with massed trials
Weakened with overtraining
Reversal of above two effects with deflation of training context
Asymmetry of posttraining inflation and deflation of overshadowing cue

Degraded contingency effect:

Basic effect (decreased responding because of outcome-alone trials)
Addition of a cover stimulus that signals the added outcomes reduces the effect.
Restoration of the effect by posttraining extinction of the cover stimulus
Signaling the target-cue–outcome trials abolishes the effect

Overexpectation effect:

Basic effect (decreased responding because of two independently trained cues being reinforced in compound)
Recovery with the extinction of one excitator

Relative stimulus-validity effect:

Basic effect (decreased responding because of training in presence of a better predictor)
Recovery with extinction of better predictor

Phenomena related to conditioned inhibition:

Existence of conditioned inhibition given different treatments (e.g., Pavlov's feature negative, explicitly unpaired)
Explicitly unpaired procedure more effective with massed trials
Feature-negative (Pavlov's) procedure more effective with spaced trials
Posttraining extinction of the training excitator decreases inhibition
Lack of extinction of conditioned inhibition with presentation of the target cue
Espinet effect (Espinet, Iraola, Bennett, & Mackintosh, 1995)

Counteraction of:

Latent inhibition and overshadowing
Degraded contingency training and overshadowing
Massed trials and overshadowing
Long conditioned stimuli and overshadowing
Blocking and blocking
Overshadowing and blocking

Superconditioning:

Existence of basic phenomenon (enhanced responding as a result of reinforcement in the presence of a conditioned inhibitor)
Extinction of the excitator used in inhibition training attenuates superconditioning

Differing changes in the associative status of cues trained together**Third-order comparator effects****Cue facilitation:**

Second-order conditioning
Sensory preconditioning
Potentiation and augmentation
Posttraining extinction of the positive mediating cue reduces excitation

Transition from second-order conditioning to Pavlovian conditioned inhibition as a function of number of XA trials

Figure 15. Phenomena modeled by sometimes-competing retrieval with a fixed set of parameters.

uli. Typically, one stimulus will come to function as the predominant first-order comparator stimulus because of its greater success in down-modulating the comparator role of the other comparator stimuli. However, the effectiveness of each first-order comparator stimulus will in turn be modulated by the associative strengths of second-order comparator stimuli (e.g., see Figure 3).

The assumption of competition among comparator stimuli distinguishes the ECH from WYSIWYH models of retrospective revaluation. All current accounts of retrospective revaluation, whether acquisition-focused or expression-focused, agree that posttraining revaluation of a first-order comparator stimulus should have an inverse effect on responding to the target cue. But only the ECH predicts that revaluation of a second-order comparator stimulus will have an effect on responding to the target cue that is in the same direction as the change in responding to the second-order comparator stimulus. Consequently, revaluation of first-order and second-order comparator stimuli should have opposite effects on responding to the target cue. In contrast, the WYSIWYH models (e.g., Dickinson & Burke, 1996; Van Hamme & Wasserman, 1994) have no mechanism for explaining such an interaction. They treat all comparator stimuli equally and therefore predict the same inverse effect on responding to any target cue when the associative status of a competing cue is altered. However, there are now appreciable published instances of second-order comparator effects that are consistent with ECH/SOCR, some of which we have described here (also see De Houwer & Beckers, 2002). In addition, the ECH does not require a direct within-compound association between the target cue and a second-order comparator stimulus for the second-order comparator stimulus to mediate responding to the target cue, as can be seen in several of the experiments we have summarized. To influence responding to a target cue, first-order comparator stimuli for $V_{X,O}$ must have a direct association with the target (X), but second-order comparator stimuli for $V_{X,O}$ need only have an association with an effective first-order comparator stimulus (e.g., $V_{A,B}$ in Figure 3). In contrast, the contemporary acquisition-focused models capable of accounting for retrospective revaluation require the manipulated cue have a direct within-compound association for there to be any influence on the behavioral control of the target cue.

Another implication of the ECH/SOCR framework is that it calls attention to the dependence of cue-competition phenomena on experimental parameters. We have reviewed a number of experimental preparations in which nominal cue-competition treatments actually attenuate otherwise existing cue competition or lead to cue facilitation. For example, for overshadowing to be observed, trials must be spaced (Stout, Chang, & Miller, 2003); cues must be of short duration (Urushihara, Stout, & Miller, 2004); and the number of compound cue–outcome pairings must be limited in number (Bellingham & Gillette, 1981; Stout, Arcediano, et al., 2003). Differential salience among two stimuli facilitates overshadowing of the less salient cue, although reciprocal overshadowing can be observed (Mackintosh, 1976). The other prime example of cue competition, blocking, is also highly dependent upon experimental parameters. For blocking to occur (A–outcome followed by AX–outcome), the experimental group must receive neither too few A–O pairings, nor too many A–O pairings (Pineño et al., 2006), nor too many AX–outcome pairings (Azorlosa & Cicala, 1988). The effect of trial spacing and cue duration has not yet been explored in this paradigm but might be expected to mimic

the parallel effects observed in overshadowing. Consistent with data from overshadowing and blocking, conditioned-inhibition treatments are also dependent on experimental parameters. The excitator must be of sufficient excitatory value (Pavlov, 1927), and, at least with the feature-negative paradigm of Pavlov (AX–nothing/A–outcome), trials must be spaced (Masiak & Frey, 1977). These examples are so numerous that merely predicting the existence of competition can no longer suffice as a benchmark for associative models; models must predict the specific parameters of training within a given protocol that are necessary for competition or facilitation to be observed. SOCR is one such model.

Shortcomings of the Model

As might be expected, the ECH fails to address some rather important phenomena in the learning literature. For example, the model has serious problems accounting for some phenomena related to experimental extinction, that is, the reduction in stimulus control of acquired responding that results from nonreinforced exposure to a cue. Within the comparator framework, one might view extinction as analogous to latent inhibition and attribute it to the extinction treatment increasing the strength of the target cue–comparator stimulus (context) association ($V_{X,j}$). An additional mechanism that contributes to SOCR's account of extinction is the unlearning mechanism for decrementing associations, represented in Equation 1B, which is also found in the models of Rescorla and Wagner (1972) and Van Hamme and Wasserman (1994). However, there is considerable evidence contrary to both of these accounts of extinction, including spontaneous recovery (Pavlov, 1927), disinhibition (Pavlov, 1927), and renewal (Bouton, 1993). Hence, although ECH/SOCR can explain the primary phenomenon of extinction, it cannot account for some important details. Although these phenomena are not readily explained through a comparator process, they are closer in spirit to a comparator view of basic learning phenomena than to a WYSIWYH view in that they bespeak retention of latent information.

Also outside the domain of SOCR are some effects of trial order. At least in some situations (primarily those with relatively short intervals between training and testing; Urushihara, Wheeler, & Miller, 2004), trial order clearly influences behavior. For example, forward blocking is usually stronger than backward blocking (i.e., less responding to cue X is observed when A–outcome pairings are followed by AX–outcome pairings than when AX–outcome pairings are followed by A–outcome pairings; e.g., Chapman, 1991). One potential explanation for this sort of trial-order effect can be cast in terms of the partial protection against backward blocking that is provided by the target cue being biologically significant at the time of the pairings of the blocking cue alone with the outcome in backward blocking treatment (Denniston, Miller, & Matute, 1996; Miller & Matute, 1996; Oberling, Bristol, Matute, & Miller, 2000). But such an account is clearly beyond the scope of ECH/SOCR. As previously stated, another potential explanation that is more compatible with ECH/SOCR is that when A–outcome trials follow AX–outcome trials, the X–A association ($V_{X,j}$) is subject to extinction. In contrast, when A–outcome trials precede the AX–outcome trials, there is no opportunity for extinction of the X–A association.

Another troublesome set of observations is that recovery from cue-competition effects can be obtained not only by posttraining

extinction of the target cue's comparator stimulus (i.e., retrospective revaluation), but also by posttraining reminder cues (e.g., Kasprow, Cacheiro, Balaz, & Miller, 1982; for a review, see Miller, Kasprow, & Schachtman, 1986) and long retention intervals (i.e., spontaneous recovery; e.g., Cole, Denniston, & Miller, 1996; Kraemer, Lariviere, & Spear, 1988; Pineño, Urushihara, & Miller, 2005). ECH/SOCR provides no account of how these treatments work to reverse cue competition. But at least the recovery of these memories is more compatible with performance-focused views of cue competition, such as the comparator hypothesis, which assume latent associations, than with WYSIWYH acquisition-focused views.

SOCR anticipates that cue competition in so-called direct learning (e.g., forward blocking [A–outcome followed by AX–outcome]) and in retrospective revaluation (e.g., backward blocking [AX–outcome followed by A–outcome]) should equally depend on the strength of the within-compound X–A association. In contradiction to this prediction, several human contingency studies have found the within-compound association to correlate far better with backward blocking than with forward blocking (e.g., Melchers et al., 2004). However, Amundson, Witnauer, Pineño, and Miller (2006), using rats in a Pavlovian preparation, recently found evidence that at least in some direct learning situations, the X–A within-compound association is critical.

A related challenge for the contention of the comparator hypothesis (and of WYSIWYH models designed to account for retrospective revaluation) that a within-compound association is required for cue competition to occur comes from several recent studies of associative interference. For example, Matute and Pineño (1998) observed retroactive interference (analogous to cue competition arising from retrospective revaluation) between cues trained apart (that is, not in compound with each other) with a common outcome. In addition, Escobar, Pineño, and Matute (2002) found that competition between cues trained apart could be alleviated through posttraining extinction of the competing cue. Moreover, not only does the occurrence of competition between cues trained apart appear to discredit the necessity of within-compound associations in promoting at least some forms of cue competition, but the size of the competition effect in the Escobar et al. study was also actually larger in the independently trained condition (but see Aitken, Larkin, & Dickinson, 2001, for evidence that in some cue-competition situations, the within-compound association is important). A common property of the aforementioned studies is that they all took place during single-session human contingency-learning experiments. It is conceivable that these situations simply obey different rules than those we have proposed to describe Pavlovian conditioning experiments with highly distributed training sessions. In the former case, the competing association may already be activated because of the recency of the relevant training trials to the test trial. Typically, in Pavlovian conditioning studies, testing takes place in a separate session, virtually requiring that competing associations be associatively retrieved. However, there are some recent reports of competition between cues trained apart in studies with highly distributed sessions in which rats were used as subjects (e.g., Amundson, Escobar, & Miller, 2003; Escobar, Matute, & Miller, 2001), so recency (i.e., absolute time, as opposed to relative time) from the competing trials to test cannot be the only factor.

A further shortcoming of SOCR and the comparator hypothesis is that it is framed in terms of associations that are strengthened and weakened over successive trials. In this sort of model, all that subjects retain are summary statistics, as opposed to memories of specific trials. There is considerable and growing evidence that nonhumans, as well as humans, store episodic memories, rather than simply impoverished summary statistics (e.g., Clayton & Dickinson, 1998; Zentall, 2005). Summary statistics are clearly an oversimplification, but we leave for the future reframing the ECH in terms of some sort of instance theory (e.g., Logan, 1988).

Conclusions

The comparator hypothesis, as well as its derivatives (such as the ECH and SOCR), as largely a response rule for the expression of acquired associative information, captures two basic truths about the nature of behavior. First, learned information is richer than a few summary statistics about target cues and outcomes. Acquired performance arises from an intricate interplay of many memories. Second, behavior is influenced by the relative values of its determinants. The cue competition, stimulus preexposure, and contingency effects discussed above are common examples of the notion of relativity in Pavlovian conditioning, as is Gibbon and Balsam's (1981) scalar expectancy theory of Pavlovian conditioning. Analogous concepts are found in the matching law (Herrnstein, 1970) of instrumental conditioning, the notion of marginal utility in motivation, economics and decision-making theory (e.g., Kahneman & Tversky, 1979), Fechner's Law in psychophysics (e.g., Garner, 1954), and optimal foraging theory (e.g., Charnov, 1976; Kamil & Roitblat, 1985). Common to all of these examples is the inverse relationship between the probability or magnitude of the target response and the base rate or value of other stimuli or responses in the situation.

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Correction to Stout and Miller (2007)

The article “Sometimes-Competing Retrieval (SOCR): A Formalization of the Comparator Hypothesis,” by Steven C. Stout and Ralph R. Miller (*Psychological Review*, Vol. 114, No. 3, pp. 759–783) contained errors.

In the right column of Table 1, the first equation currently reads as follows:

$$\Delta V_{X,O} = \alpha_X * \beta_O(\lambda_O \Sigma - V_j).$$

It should read as follows:

$$\Delta V_{X,O} = \alpha_X * \beta_O(\lambda_O - \Sigma V_j).$$

In the right column of Table 1, the 11th equation currently reads as follows:

$$\Delta Op_{X,j,O} = X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}), \text{ when } V_{X,O} = 0.$$

It should read as follows:

$$\Delta Op_{X,j,O} = \alpha_X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}), \text{ when } V_{X,O} = 0.$$

Equation 6B currently reads as follows:

$$\Delta Op_{X,j,O} = X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}), \text{ when } V_{X,O} = 0.$$

It should read as follows:

$$\Delta Op_{X,j,O} = \alpha_X * k3 * V_{X,j} * V_{j,O} * (1.0 - Op_{X,j,O}), \text{ when } V_{X,O} = 0.$$